Integração com IIoT

Professor: Roberto

Aula 06

Cronograma Teórico

01 Introdução ao MQTT: Plataforma Blynk.iot

02 Desafios

01

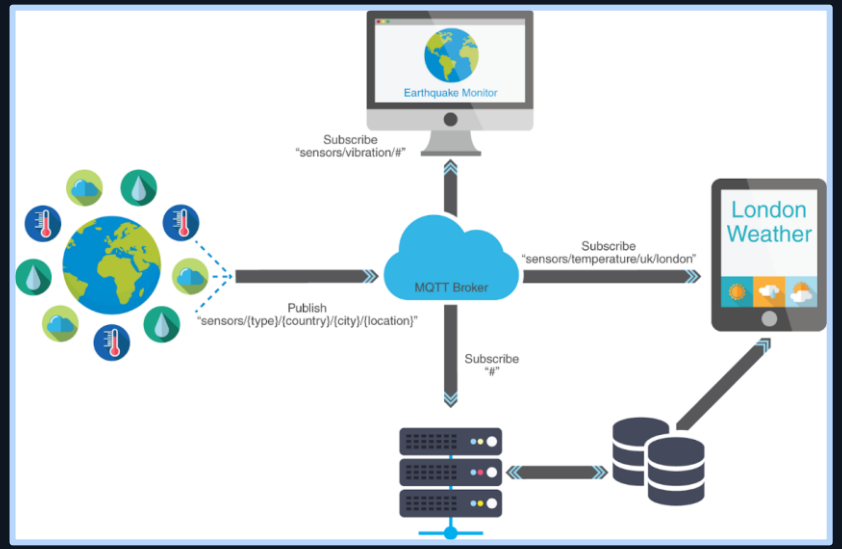
Revisão ao MQTT

MQTT

Qual é a função de um broker no MQTT?

Dê um exemplo de um tópico hierárquico.

Quando você usaria QoS 0 versus QoS 2?



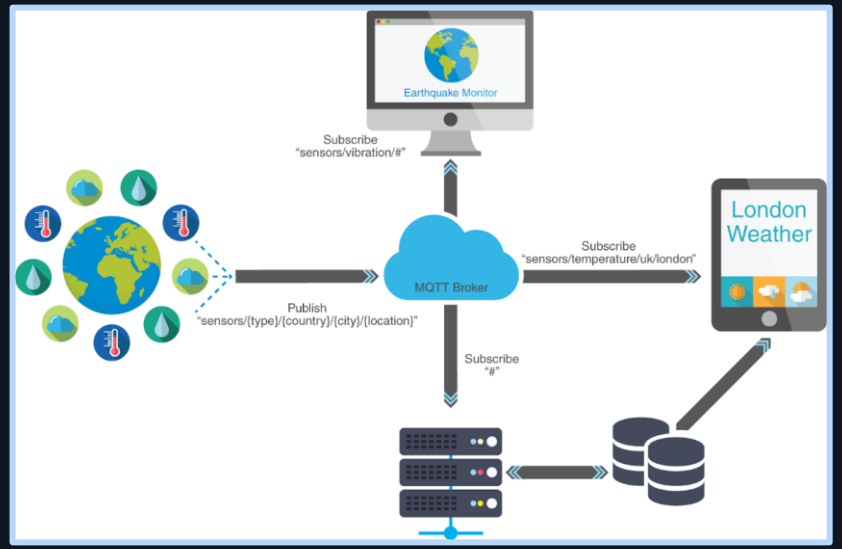
MQTT

Qual é a função de um broker no MQTT?

Protocolo leve para comunicação IoT.
Baseado no modelo publish/subscribe.
Envia e recebe menagens pela Internet.

Ideal para:

- Dispositivos com recursos limitados.
- Redes com largura de banda restrita.



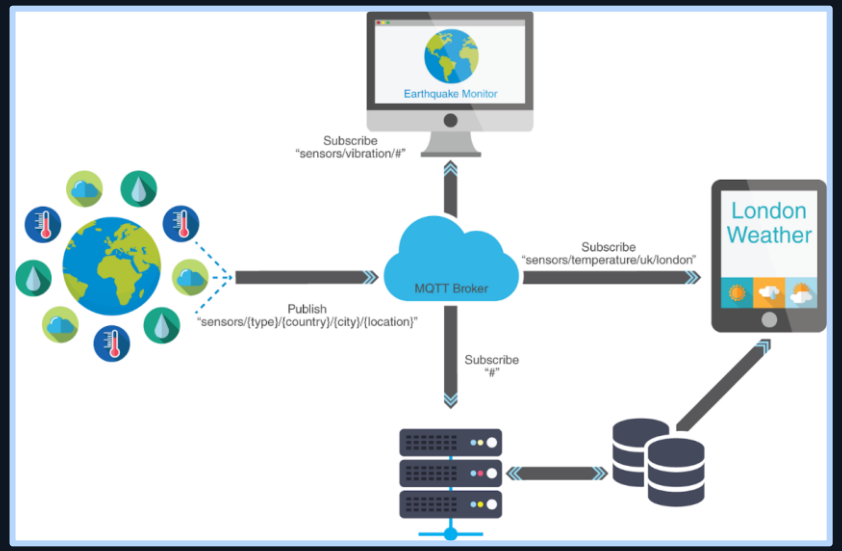
MQTT

Componentes Principais:

- Broker: Gerencia mensagens e tópicos.
- Publisher: Publica mensagens em tópicos específicos.
- Subscriber: Recebe mensagens ao se inscrever em tópicos.

Fluxo Básico:

Publisher envia mensagens para o broker.
Broker distribui mensagens para subscribers inscritos.
Mensagens são organizadas por tópicos hierárquicos.



MQTT

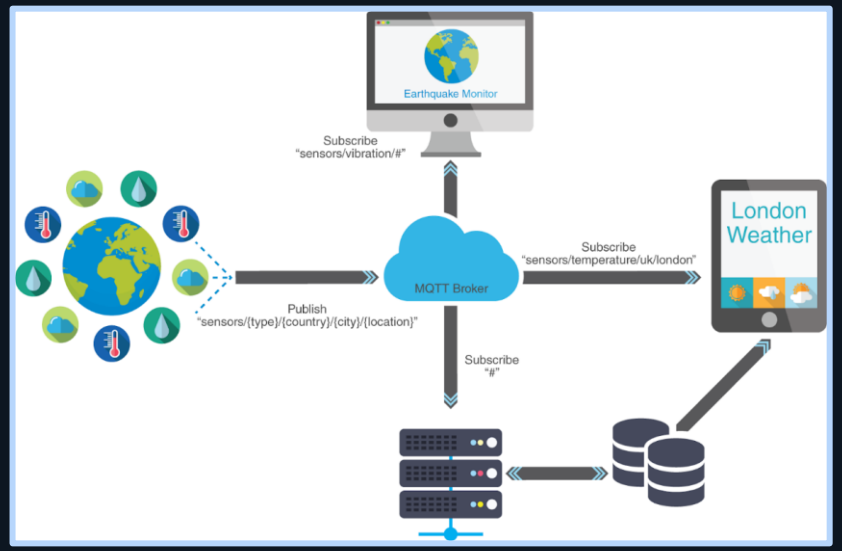
Tópicos no MQTT
Caminhos hierárquicos usados para organizar mensagens.

Exemplo:

- casa/luz/sala (local específico).
- sensor/temperatura/cozinha (dado de um sensor).

QoS (Quality of Service):

- **QoS 0:** Entrega sem garantias, ideal para dados não críticos.
- **QoS 1:** Entrega pelo menos uma vez, mas pode haver duplicatas.
- **QoS 2:** Entrega única garantida, usado para dados críticos.

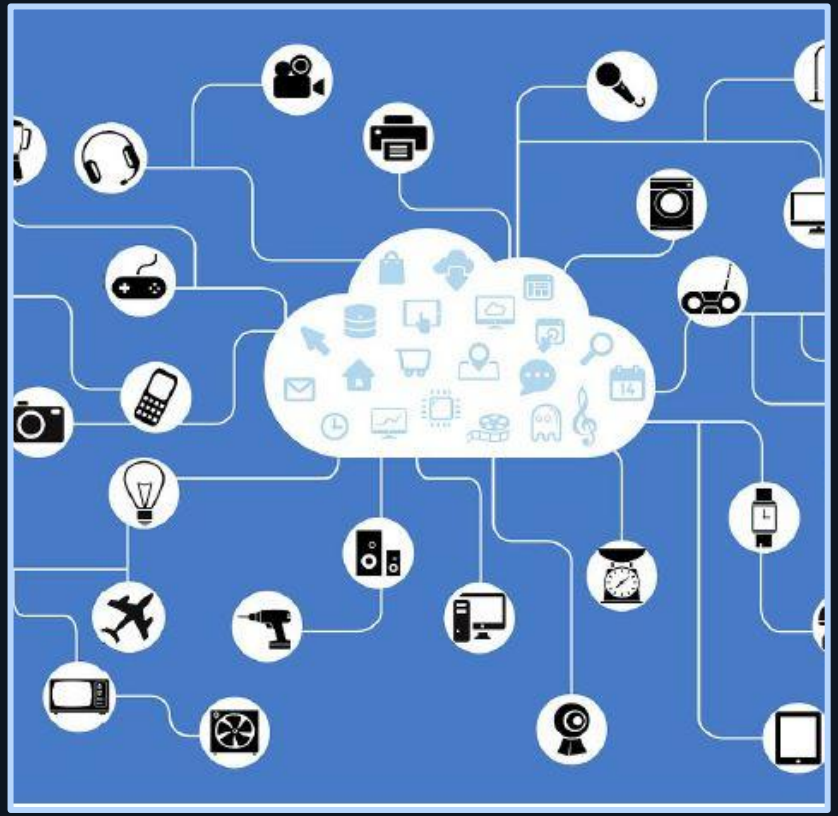


Plataforma Profissionais

Plataformas de MQTT oferecem serviços avançados para gerenciar a comunicação de dispositivos IoT, incluindo brokers, segurança e análise de dados.

Por que usar?

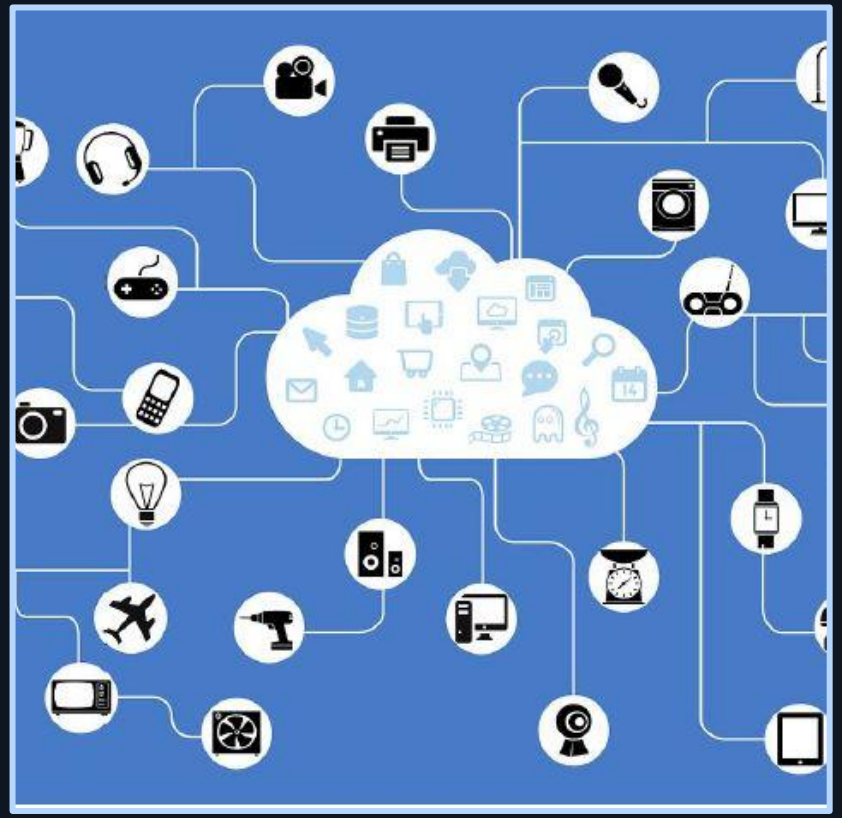
- Escalabilidade para milhares de dispositivos.
- Recursos de segurança integrados.
- Ferramentas para monitoramento e análise em tempo real.

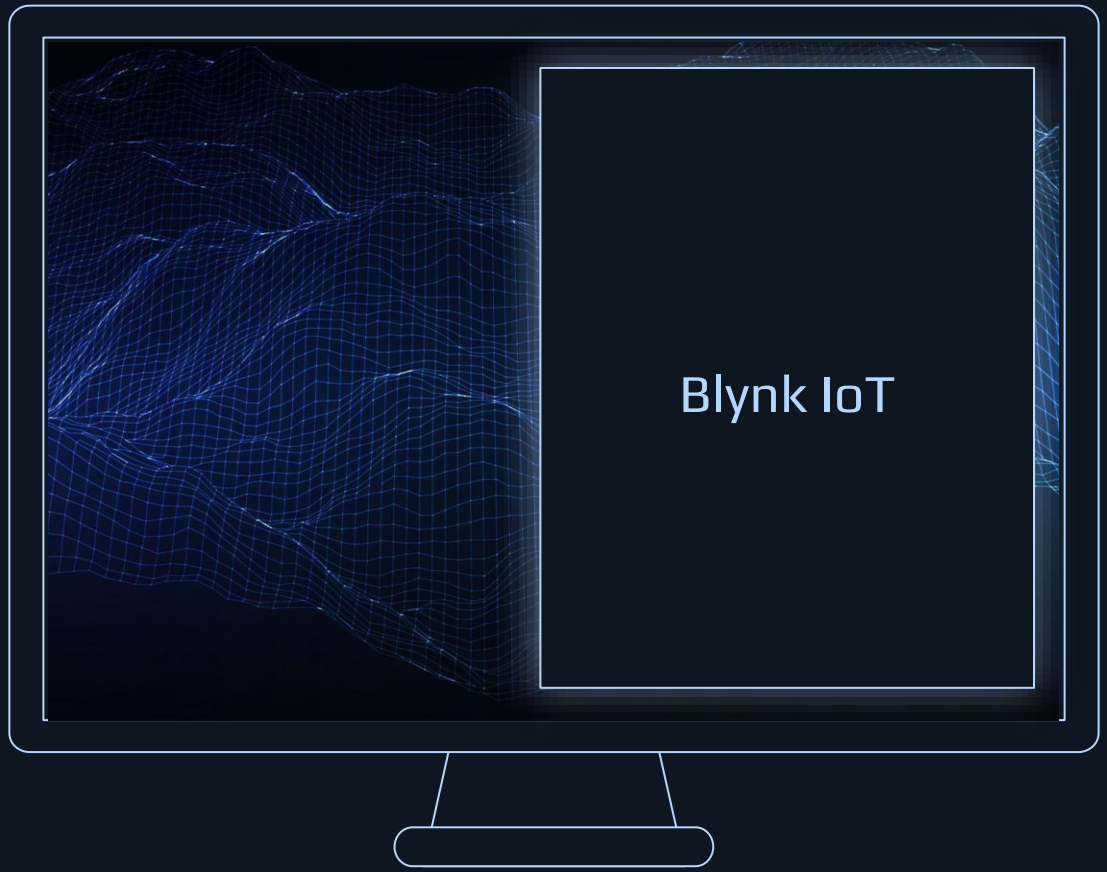


Plataforma Profissionais

Exemplos de plataformas:

- AWS IoT Core;
- HiveMQ;
- Adafruit IO;
- ThingsBoard;
- Google Cloud IoT Core;
- Azure IoT Hub;
- WMQX;
- Ubidots;
- Losant;
- Kaa IoT;
- Watson IoT Plataform;
- Particle;
- ThingSpeak
- TagoIO




Blynk IoT

Blynk IoT

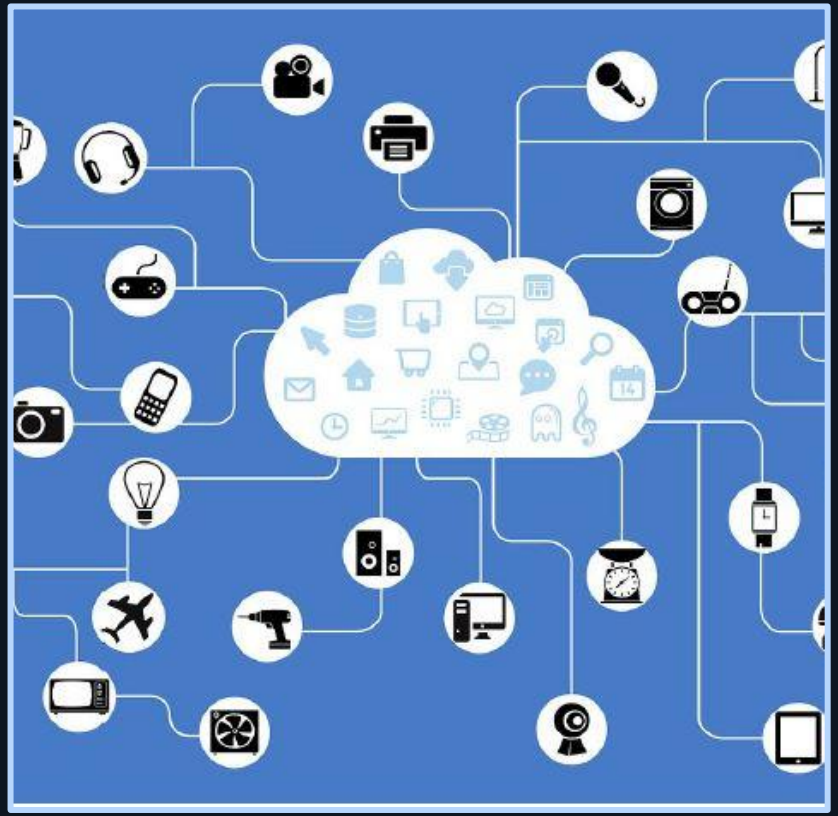
Blynk IoT é uma plataforma que permite controlar dispositivos IoT e visualizar dados por meio de aplicativos móveis ou web.

Características:

- Controle remoto de dispositivos.
- Criação de dashboards interativos.
- Suporte a protocolos como MQTT e HTTP.

Aplicações Reais:

- Automação residencial.
- Monitoramento ambiental.
- Projetos educacionais e acadêmicos.

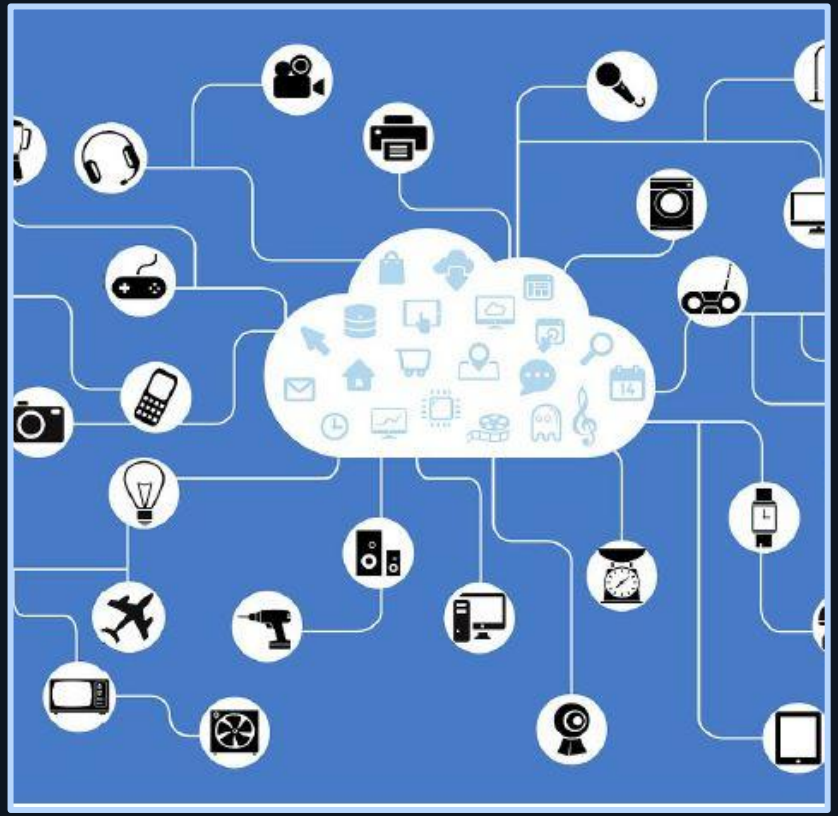


Blynk IoT - Funcionamento

Dispositivos: Microcontroladores como ESP8266 ou ESP32.
Conexão: Dispositivos se conectam ao Blynk via internet.
Dashboard: Interface gráfica para controlar e visualizar dados.

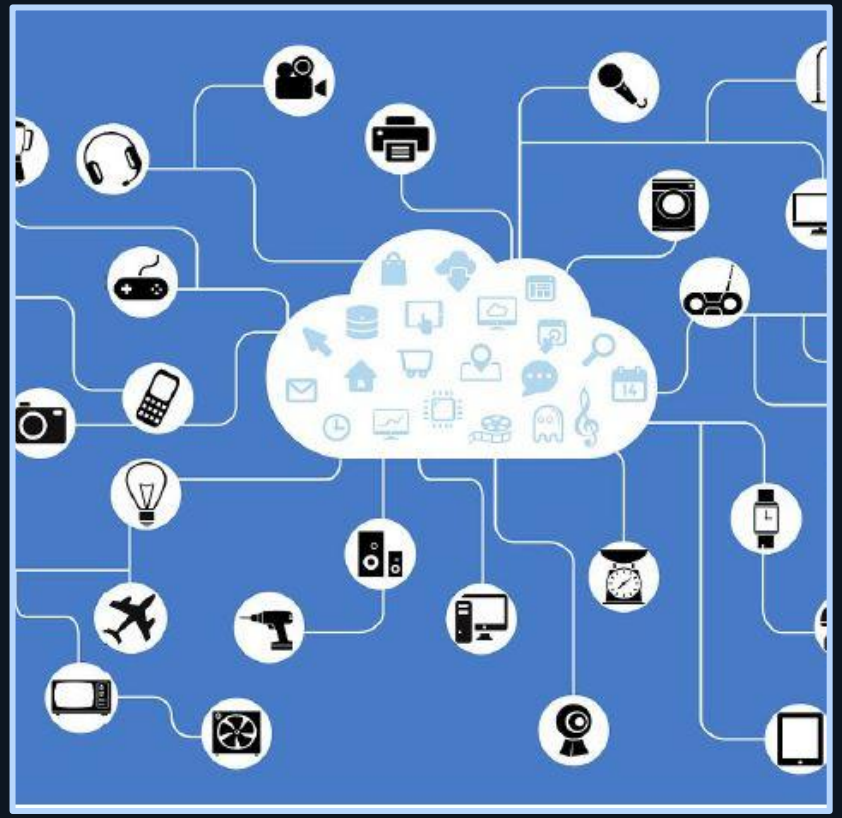
Fluxo Geral:

- Configuração do dispositivo.
- Envio/recebimento de dados para o Blynk.
- Visualização e controle via dashboard.

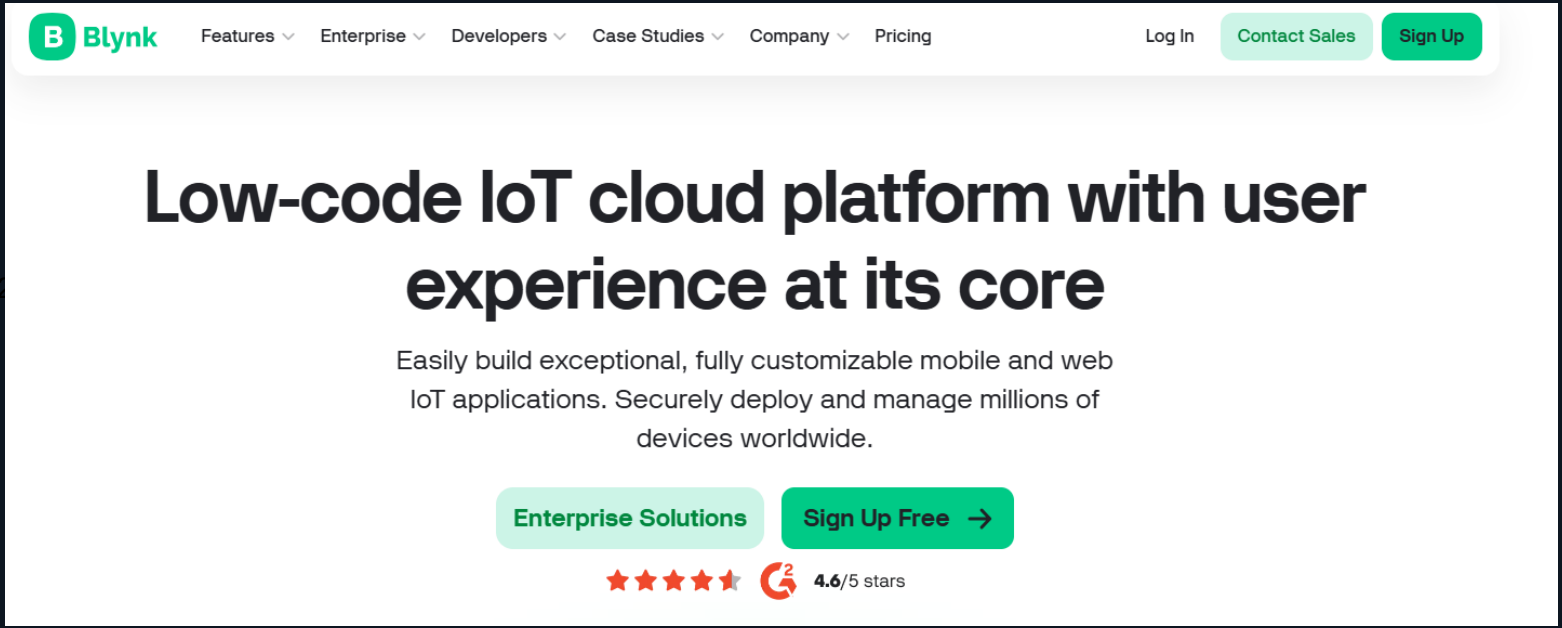


Blynk IoT - Benefícios

- Interface intuitiva e fácil de usar.
- Suporte para dispositivos populares como ESP8266, ESP32 e Raspberry Pi.
- Integração com serviços de nuvem.
- Planos gratuitos disponíveis para testes e pequenos projetos.



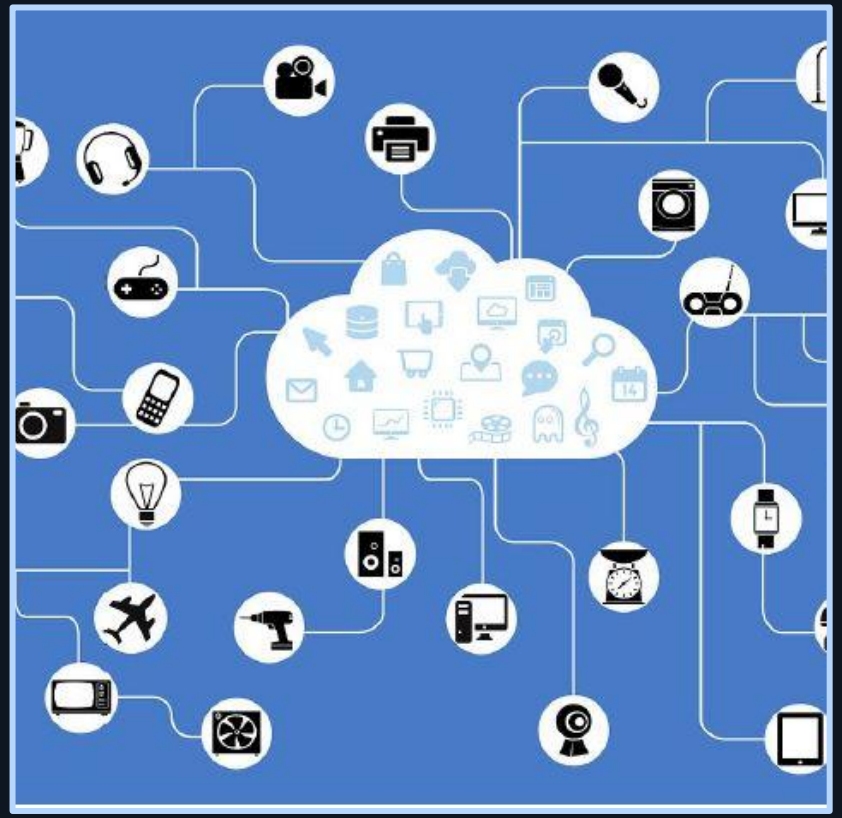
Tela Blynk



ESP82

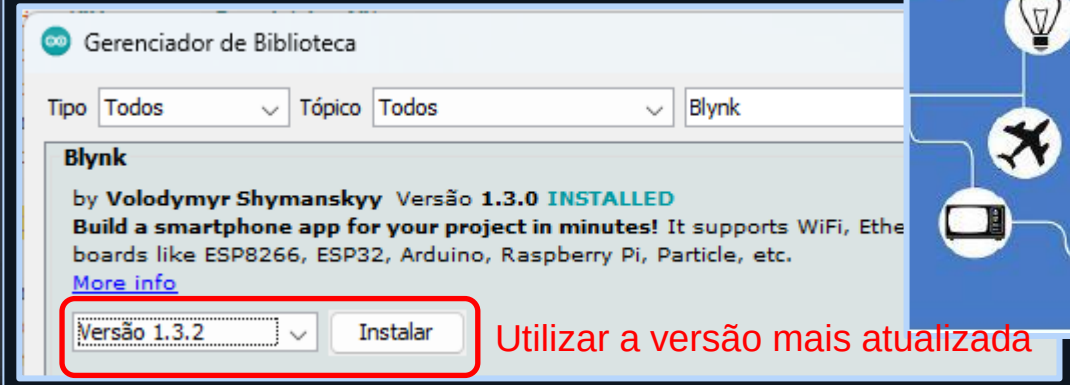
Blynk IoT - Configuração

1. Crie uma conta em Blynk.
2. Configure um novo projeto.
3. Escolha o dispositivo (ex.: ESP8266) e o método de conexão (Wi-Fi).
4. Receba o token de autenticação por *e-mail.

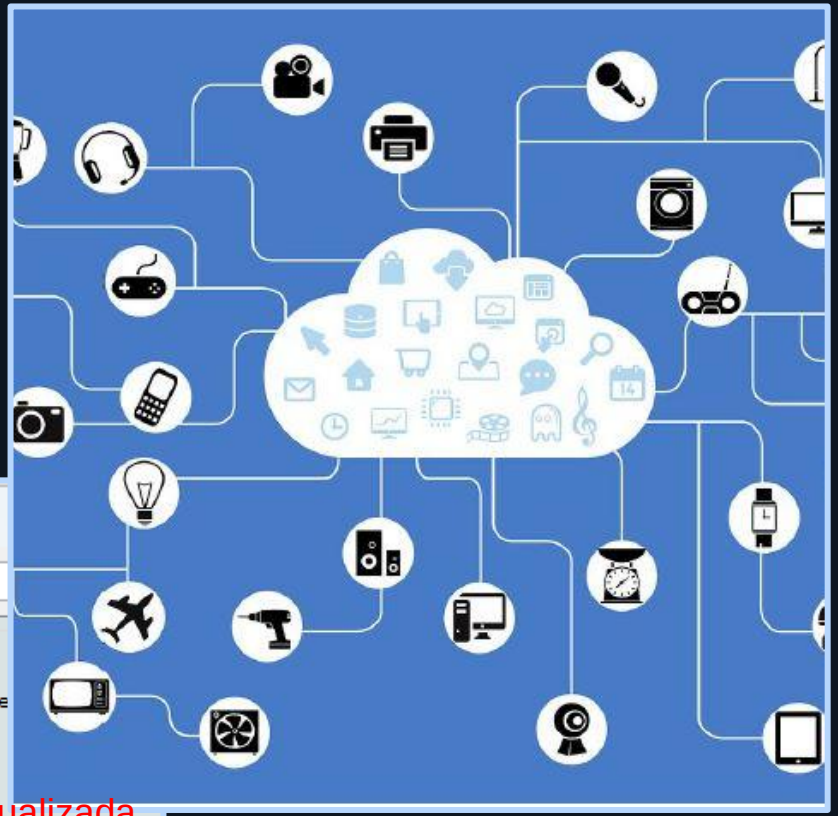


Blynk IoT

1. Para a configuração da IDE, para a utilização do Blynk é simples.
2. Devemos instalar a biblioteca Blynk na IDE
   1. Sketch → Incluir Biblioteca → Gerenciar Bibliotecas.



Utilizar a versão mais atualizada

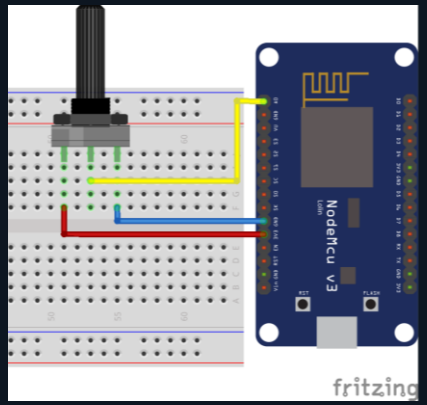


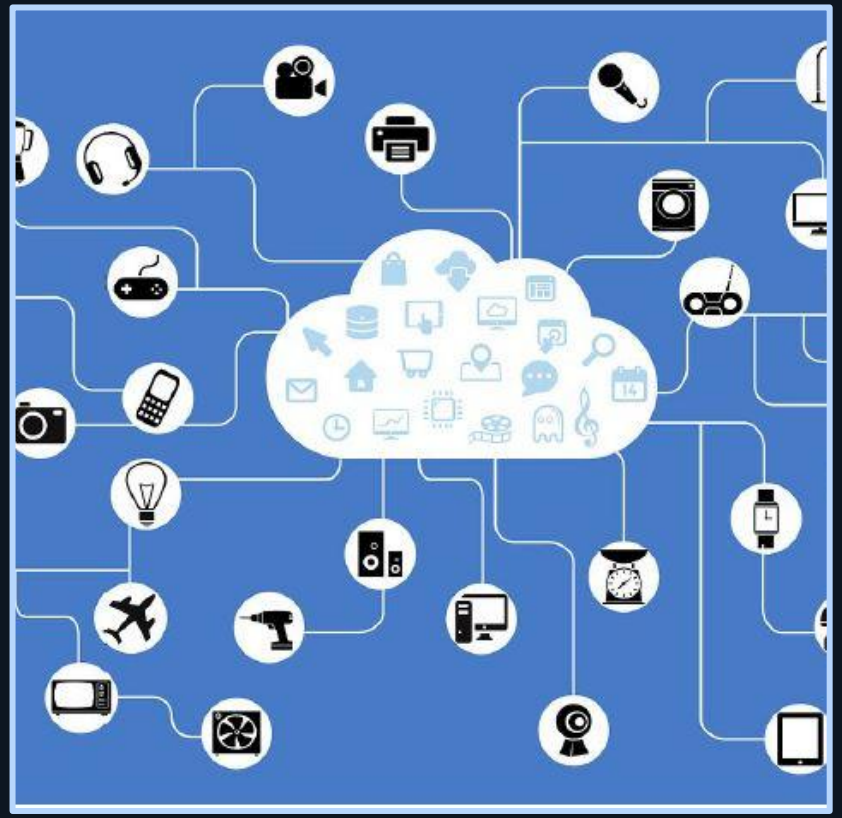
Blynk

Vamos abrir o exemplo do código em:

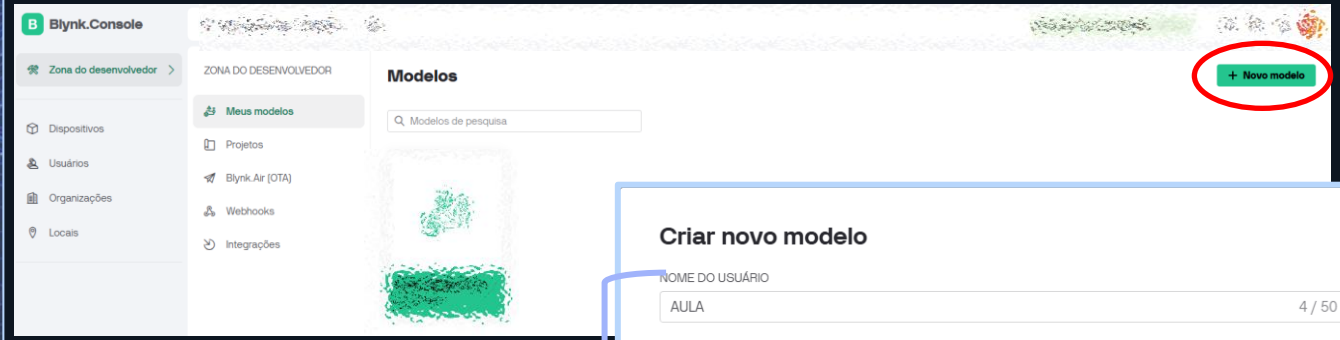
Arquivo → Exemplos → Blynk → Boards WiFI → ESP8266_standalone

Vamos criar o primeiro programa: Vamos mandar um sinal analógico para o blynk através de um potenciômetro.

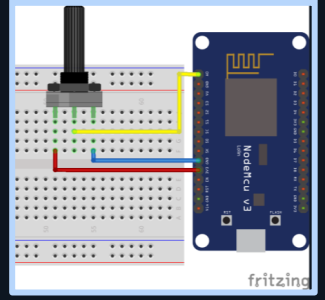




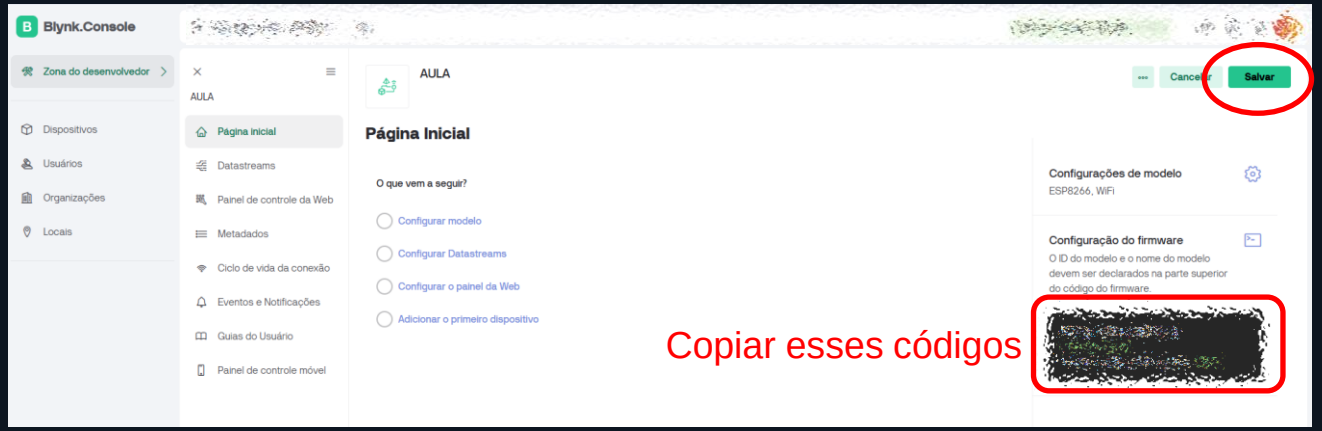
Blynk



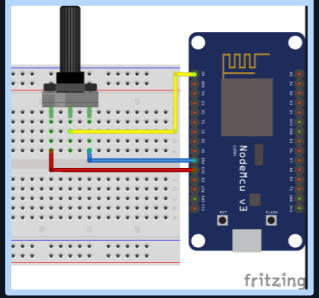
Nome do usuário: AULA
Hardware: ESP8266
Conexão: WiFi



Blynk

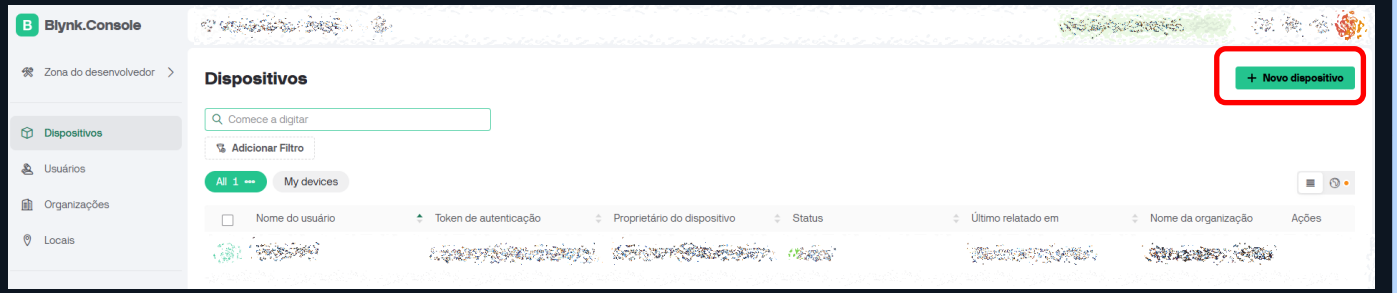


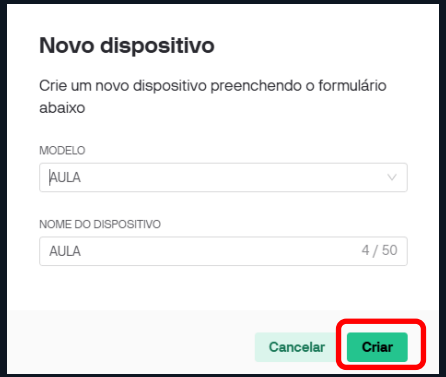
Copiar esses códigos

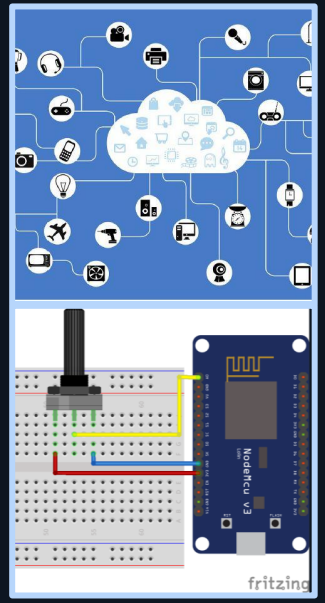


Blynk

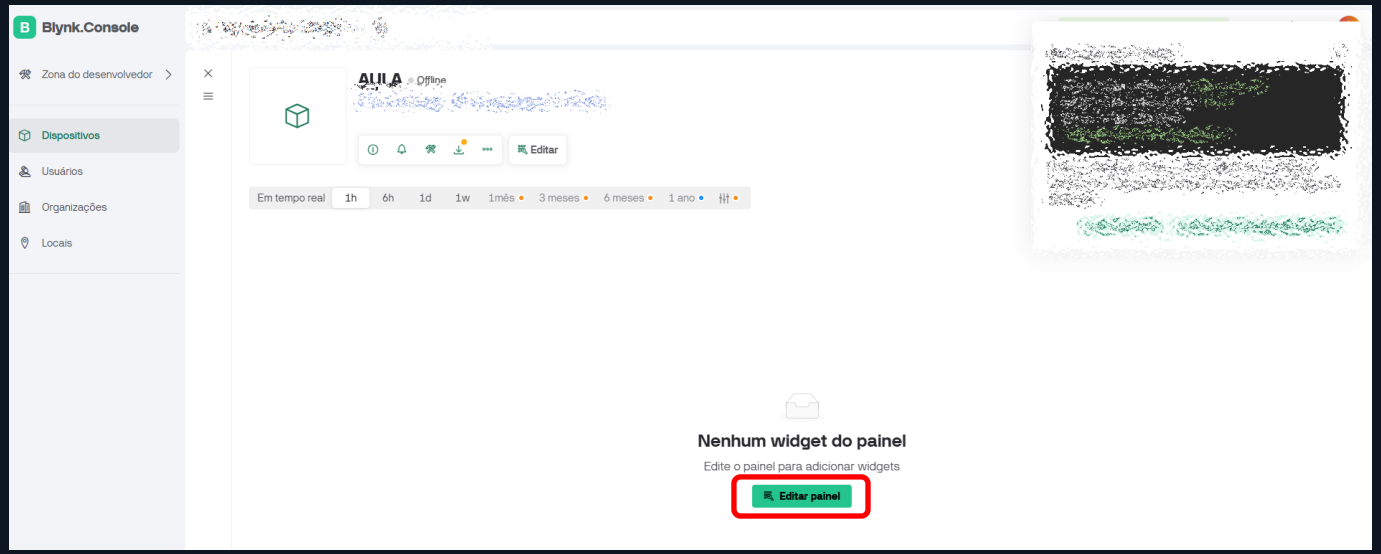
Vamos configurar o Dispositivo:

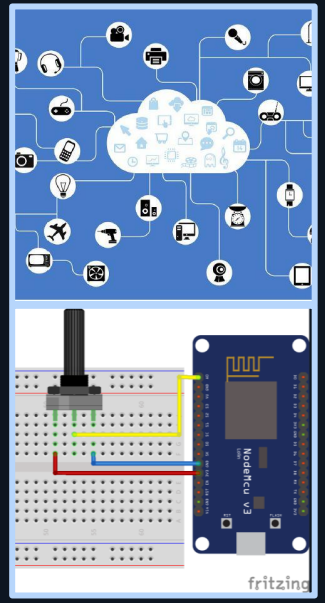




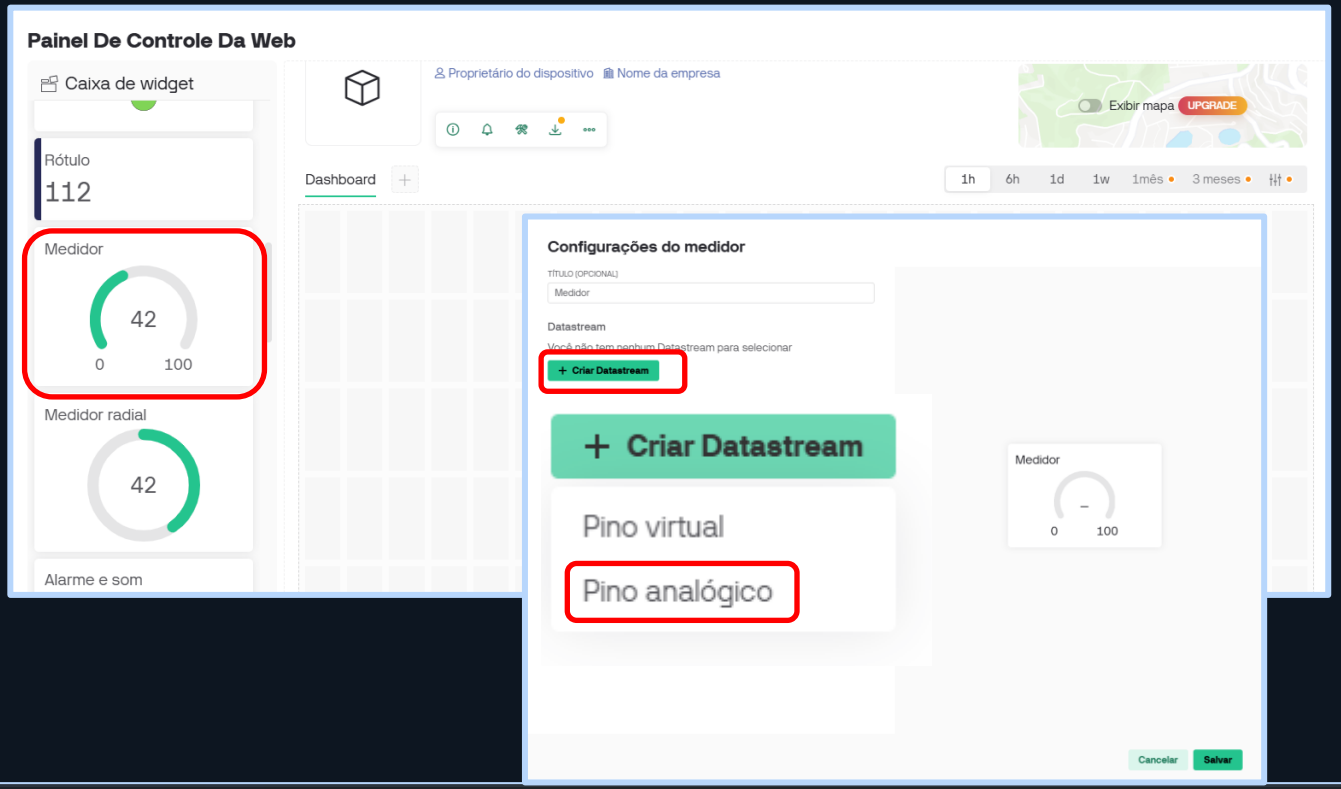


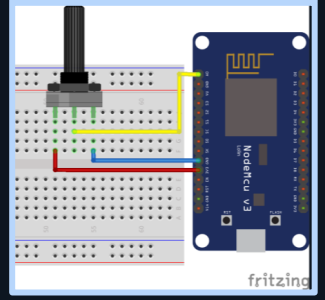
Blynk


Blynk.Console
Zona do desenvolvedor
Dispositivos
Usuários
Organizações
Locais
ALILA
Offline
Editar
Em tempo real
1h
6h
1d
1w
1mês
3 meses
6 meses
1 ano
Nenhum widget do painel
Edite o painel para adicionar widgets
Editar painel

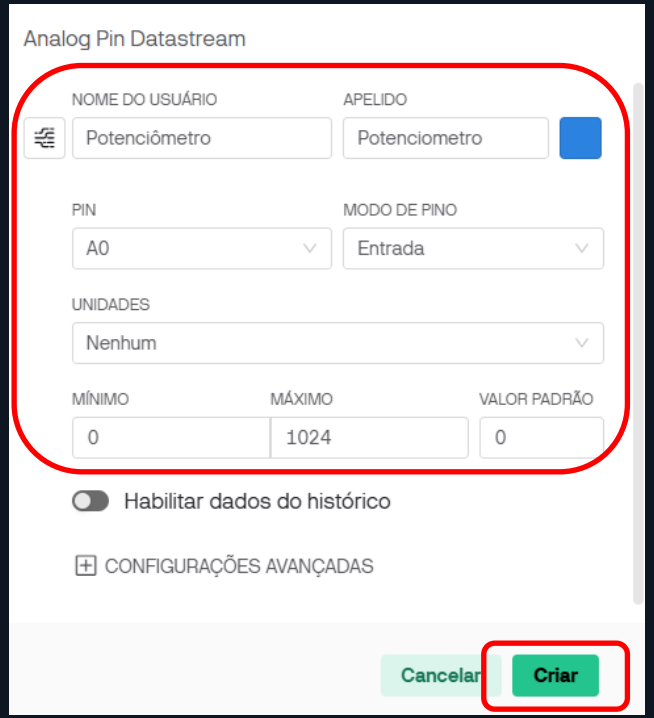

NodeMcu v3
fritzing

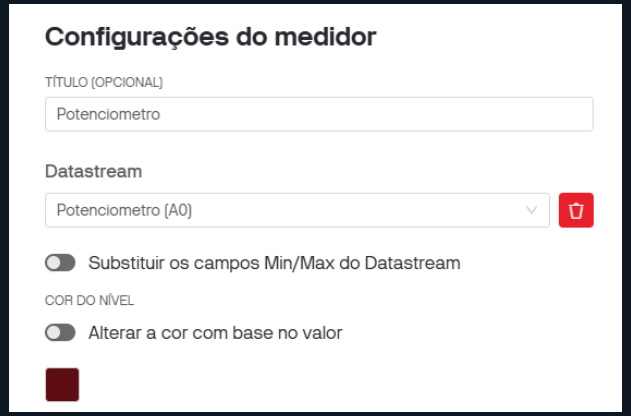
Blynk

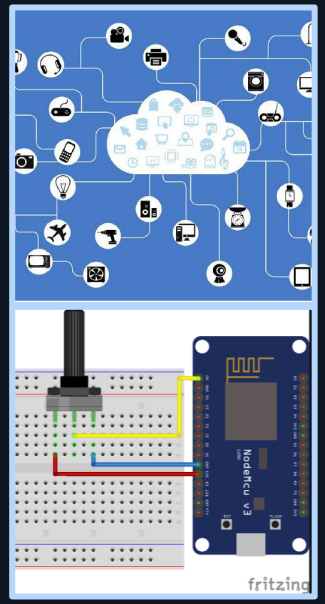

Painel De Controle Da Web
Caixa de widget
Rótulo
112
Medidor
42
0
100
Medidor radial
42
Alarme e som
Proprietário do dispositivo
Nome da empresa
Exibir mapa
UPGRADE
Dashboard
1h
6h
1d
1w
1mês
3 meses
Configurações do medidor
TÍTULO (OPCIONAL)
Medidor
Datastream
Você não tem nenhum Datastream para selecionar
+ Criar Datastream
+ Criar Datastream
Pino virtual
Pino analógico
Medidor
0
100
Cancelar
Salvar


NodeMcu V3
fritzing

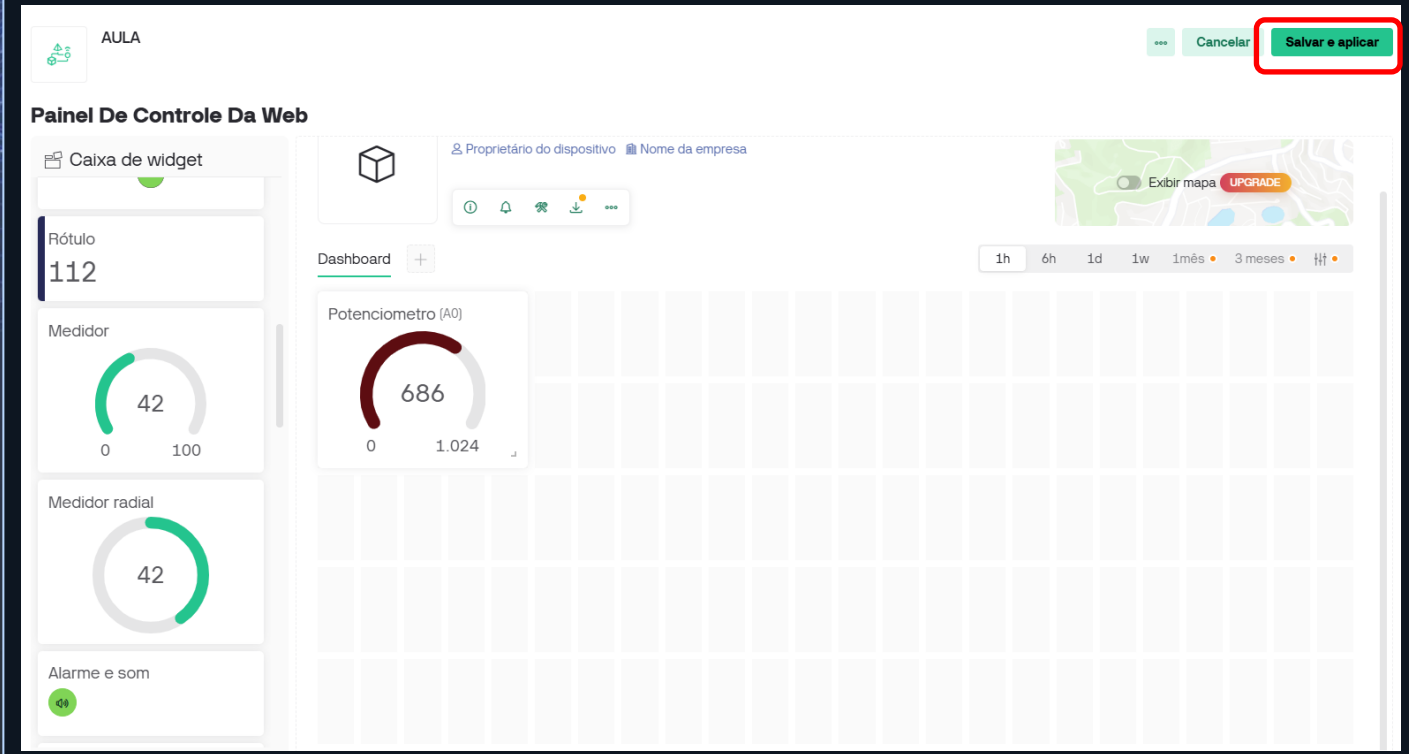
Blynk

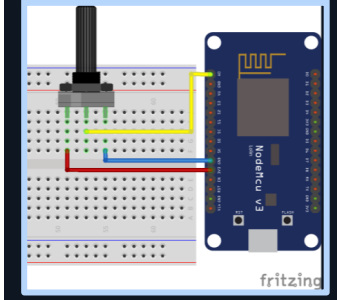

Analog Pin Datastream
NOME DO USUÁRIO
APELIDO
Potenciômetro
Potenciometro
PIN
MODO DE PINO
A0
Entrada
UNIDADES
Nenhum
MÍNIMO
MÁXIMO
VALOR PADRÃO
0
1024
0
Habilitar dados do histórico
CONFIGURAÇÕES AVANÇADAS
Cancelar
Criar


Configurações do medidor
TÍTULO (OPCIONAL)
Potenciometro
Datastream
Potenciometro (A0)
Substituir os campos Min/Max do Datastream
COR DO NÍVEL
Alterar a cor com base no valor

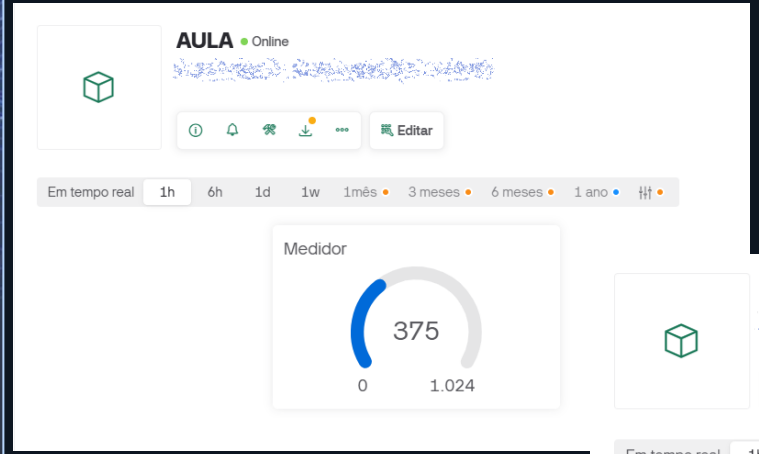

fritzing

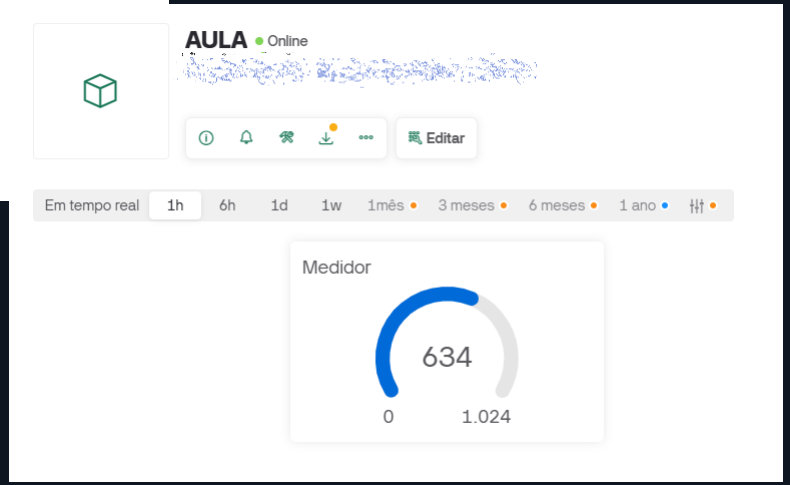
Blynk

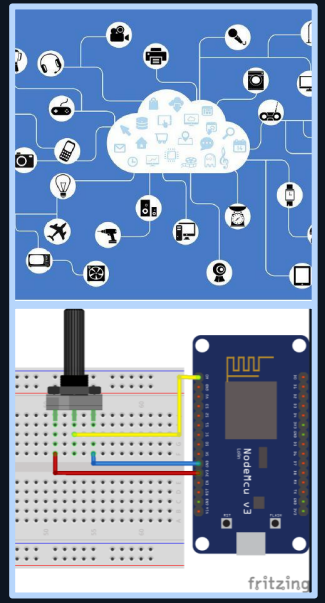

AULA
Cancelar
Salvar e aplicar
Painel De Controle Da Web
Caixa de widget
Rótulo
112
Medidor
42
0
100
Medidor radial
42
Alarme e som
Proprietário do dispositivo
Nome da empresa
Exibir mapa
UPGRADE
Dashboard
1h
6h
1d
1w
1mês
3 meses
Potenciometro (A0)
686
0
1.024


NodeMcu V3
fritzing

Blynk

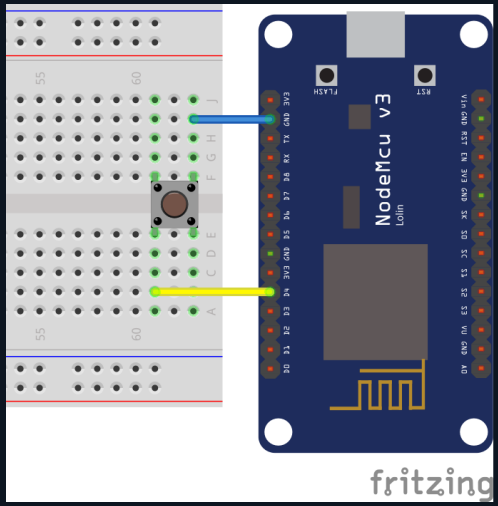

AULA
Online
Editar
Em tempo real
1h
6h
1d
1w
1mês
3 meses
6 meses
1 ano
Medidor
375
0
1.024

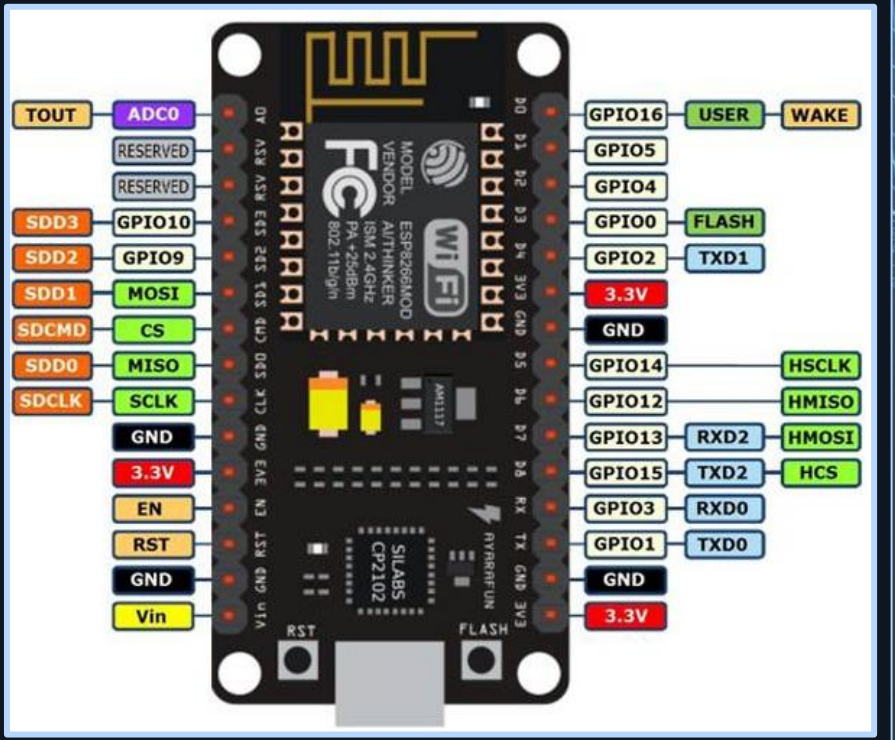

AULA
Online
Editar
Em tempo real
1h
6h
1d
1w
1mês
3 meses
6 meses
1 ano
Medidor
634
0
1.024


NodeMcu V3
fritzing

Blynk

Faça um programa que a hora que pressionar um botão ligue um LED virtual, pressionou novamente o LED desliga.





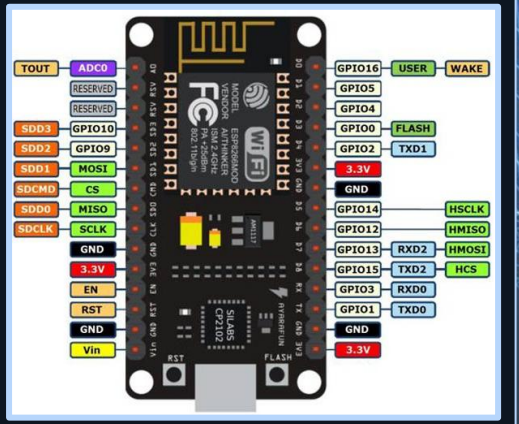
Blynk

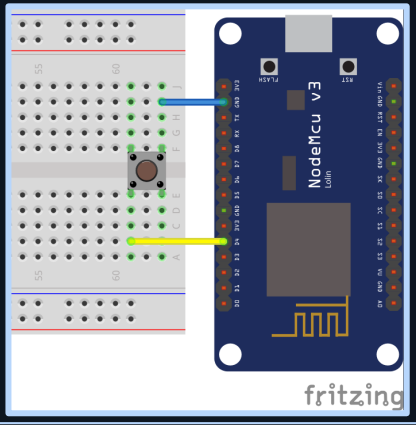
Para podermos fazer isso vamos entender o conceito de pino Virtual.

Pinos virtuais são interfaces lógicas que permitem a comunicação entre widgets no Blynk e o código do microcontrolador.

Por que usar?

- Separação entre lógica de hardware e lógica de aplicação.
- Controle avançado e flexível sobre widgets e dispositivos.

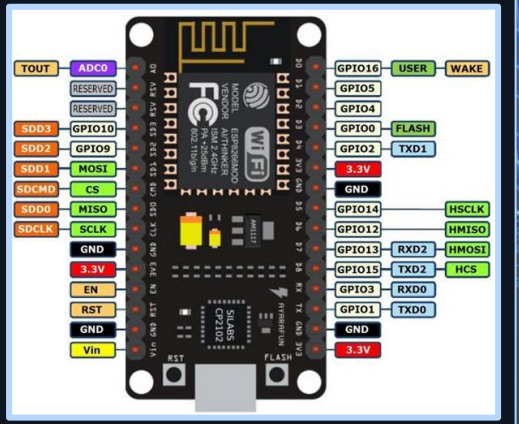


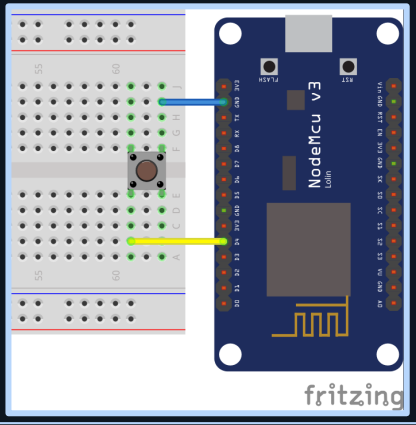


Blynk

Vantagens dos Pinos Virtuais:

- **Personalização:**
  - Controle completo sobre como os dados são processados e exibidos.
- **Integração:**
  - Facilita o uso de cálculos, condições e manipulação de dados antes de enviá-los ou exibi-los.
- **Flexibilidade:**
  - Conecte múltiplos widgets a um único pino virtual.
  - Realize tarefas que pinos físicos não permitem.

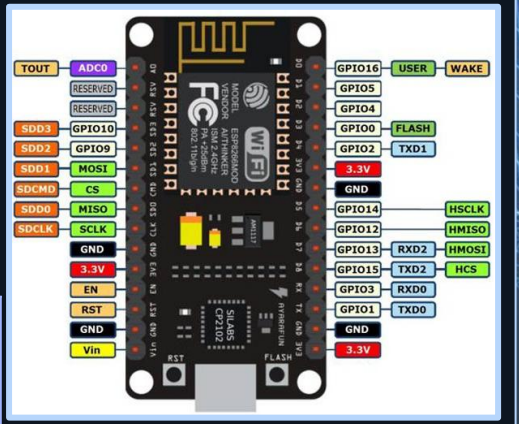


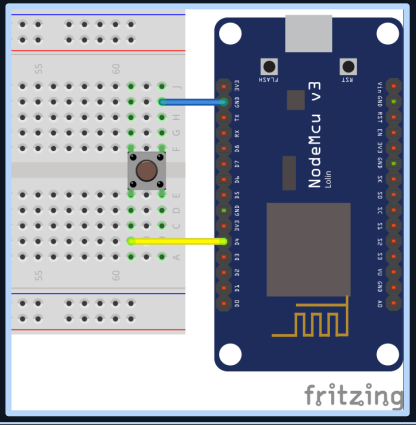


Blynk

Diferenças entre Pinos Virtuais e Físicos:

	Pino Virtual	Pino Físico
Definição	Lógica	Hardware
Função	Comunicação e controle Avançado	Controle direto de GPIOs
Exemplo de Uso	Dashboard dinâmicos	Ligar/desligar LEDs



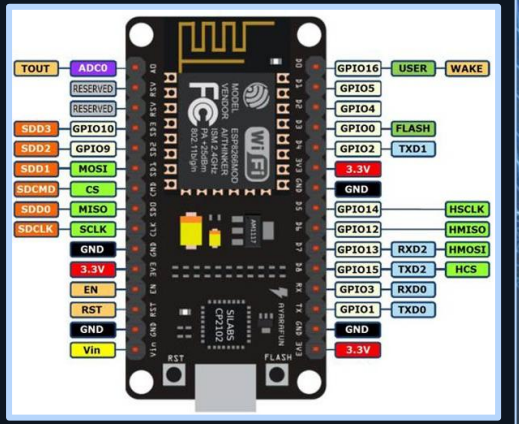


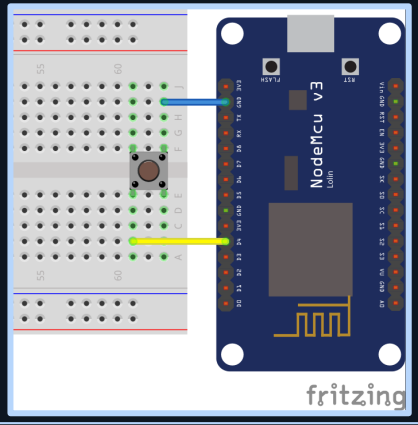
Blynk

Como Configurar um Pino Virtual no Blynk

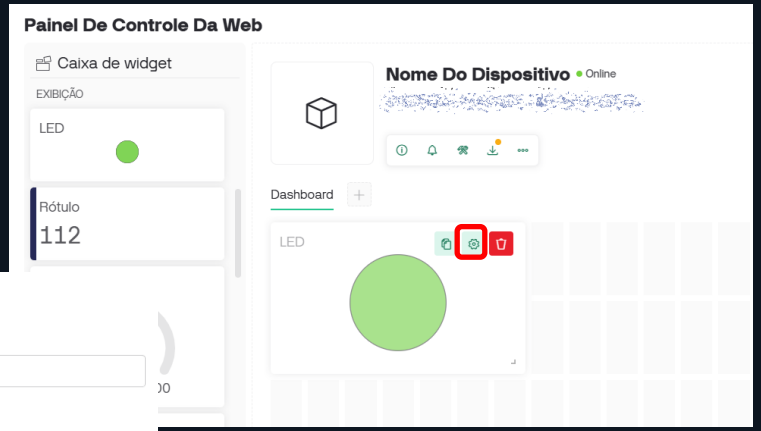
- **No Aplicativo:**
  - Adicione um widget (ex.: botão, slider, gráfico).
  - Escolha o tipo de pino como "Virtual".
  - Atribua um número ao pino (ex.: V0, V1, etc.).

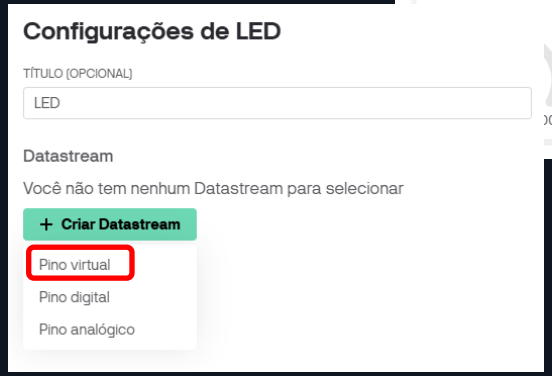
- **No Código:**
  - Use funções específicas da biblioteca Blynk:
  - **BLYNK_WRITE():** Ler dados recebidos no pino virtual.
  - **Blynk.virtualWrite():** Enviar dados para o pino virtual.

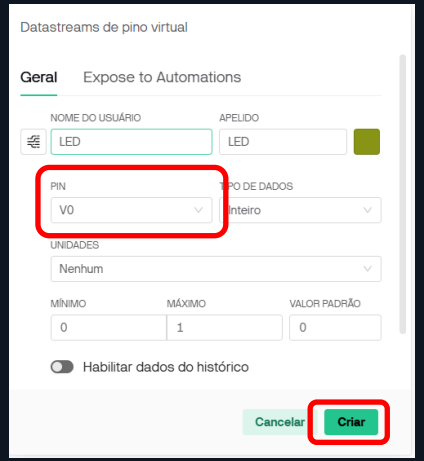


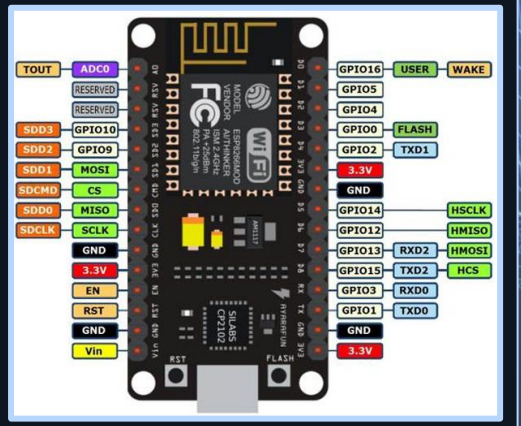


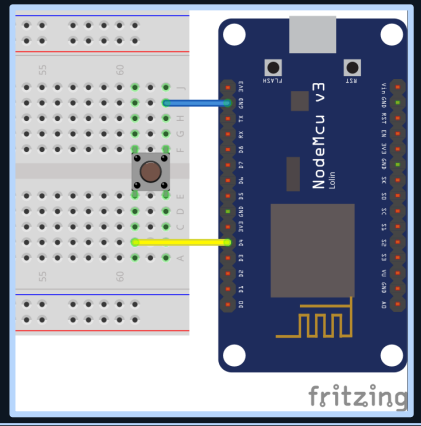
Blynk


Painel De Controle Da Web
Caixa de widget
EXIBIÇÃO
LED
Rótulo
112
Nome Do Dispositivo
Online
Dashboard
LED


Configurações de LED
TÍTULO (OPCIONAL)
LED
Datastream
Você não tem nenhum Datastream para selecionar
+ Criar Datastream
Pino virtual
Pino digital
Pino analógico

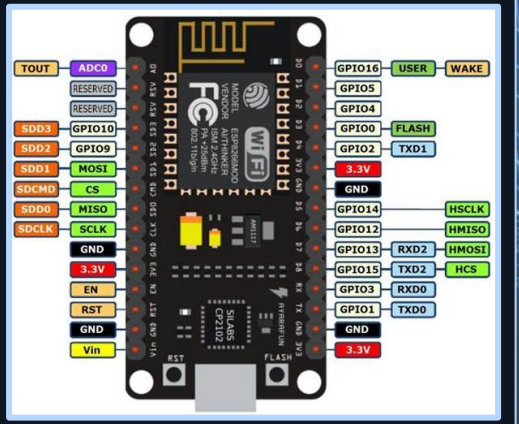

Datastreams de pino virtual
Geral
Expose to Automations
NOME DO USUÁRIO
LED
APELIDO
LED
PIN
V0
TIPO DE DADOS
Inteiro
UNIDADES
Nenhum
MÍNIMO
0
MÁXIMO
1
VALOR PADRÃO
0
Habilitar dados do histórico
Cancelar
Criar

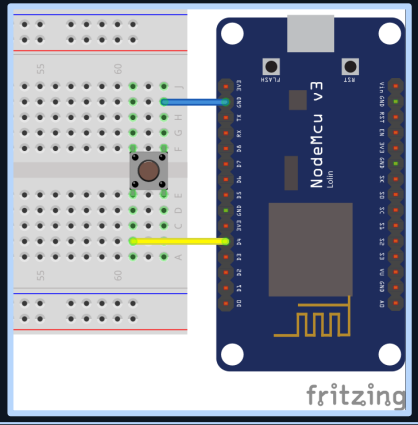

TOUT
ADC0
RESERVED
RESERVED
SDD3
GPIO10
SDD2
GPIO9
SDD1
MOSI
SDCMD
CS
SDD0
MISO
SDCLK
SCLK
GND
3.3V
EN
RST
GND
Vin
GPIO16
USER
WAKE
GPIO5
GPIO4
GPIO0
FLASH
GPIO2
TXD1
3.3V
GND
GPIO14
HSCLK
GPIO12
HMISO
GPIO13
RXD2
HMOSI
GPIO15
TXD2
HCS
GPIO3
RXD0
GPIO1
TXD0
GND
3.3V


NodeMcu v3
fritzing

Blynk

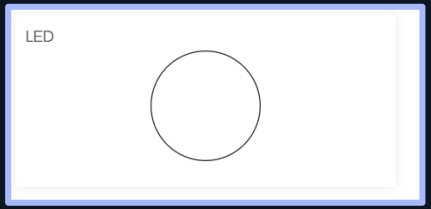
```
#define BLYNK_PRINT Serial
#define BLYNK_TEMPLATE_ID “TEMP"
#define BLYNK_TEMPLATE_NAME “NAME"
#define BLYNK_AUTH_TOKEN “TK"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

bool estado;
int btn = 2;
unsigned long tempodebounce;

char ssid[] = “SSID";
char pass[] = “SENHA";
```

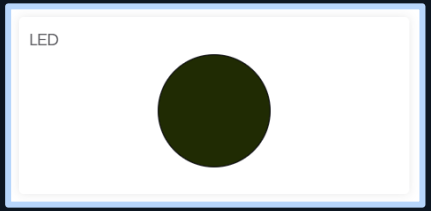


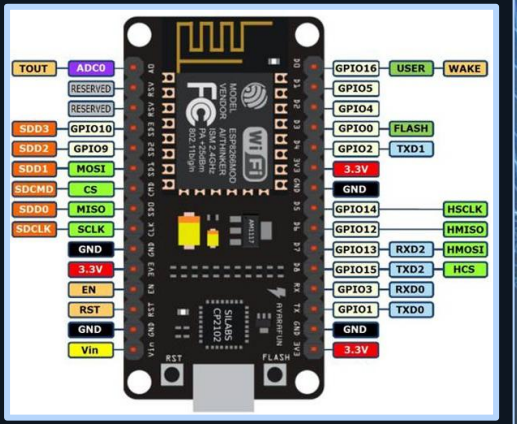


Blynk

```
void setup()
{
  Serial.begin(9600);
  pinMode(btn, INPUT_PULLUP);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
}

void loop()
{
  if (digitalRead(btn)==0){
    if ((millis() - tempodebounce)> 150){
      estado = !estado;
      Blynk.virtualWrite(V0,estado);
      Serial.println(estado);
    }
    tempodebounce = millis();
  }
  Blynk.run();
}
```

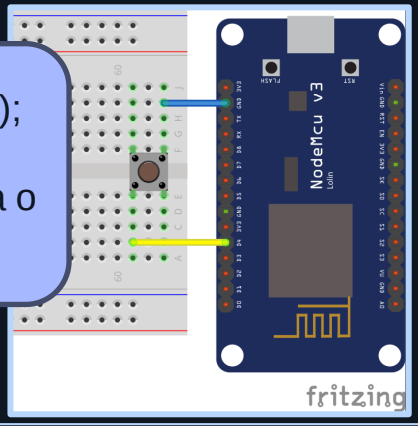






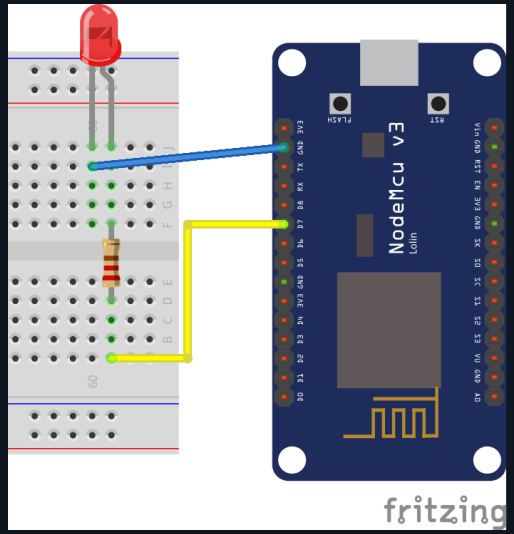
Blynk.virtualWrite(pinovirtual, Valor);

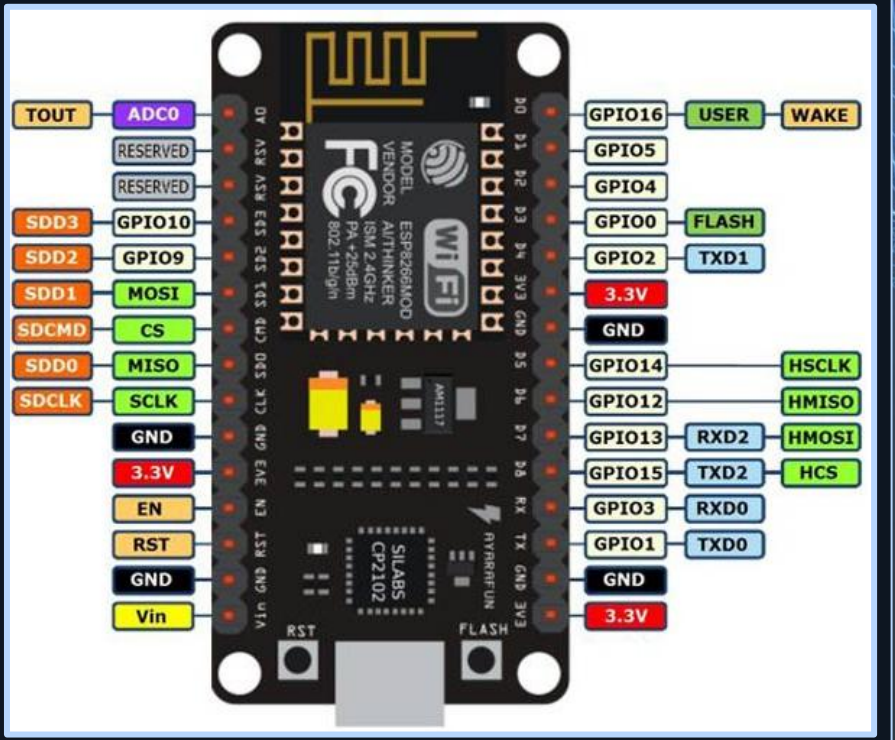
Podemos enviar qualquer valor para o pino Virtual.



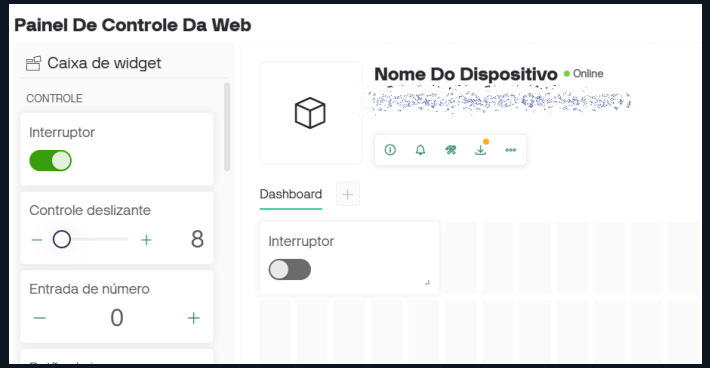
Blynk

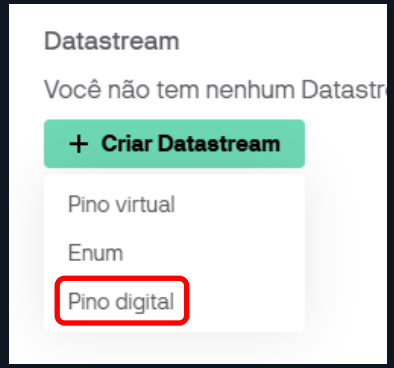
Faça um código onde um botão virtual ligue e desligue um LED físico.
Vamos fazer de 2 modos.

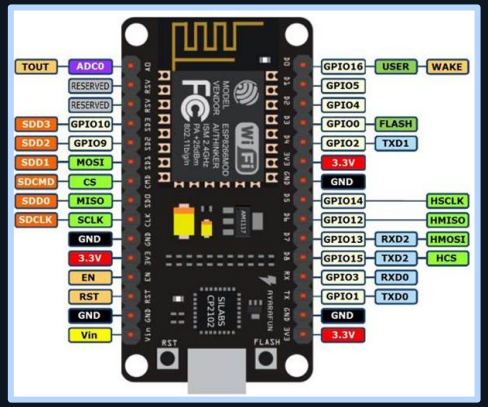


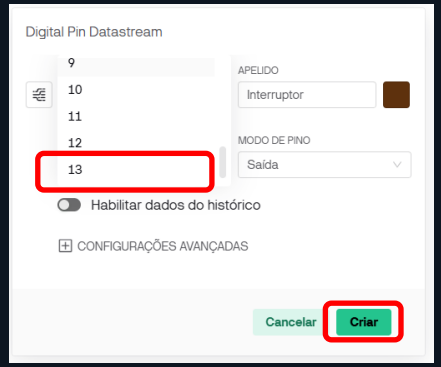


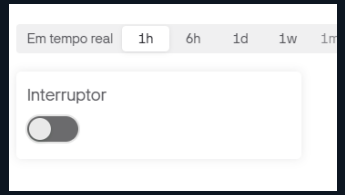
Blynk – Pino Digital

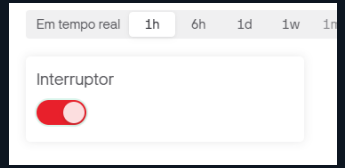


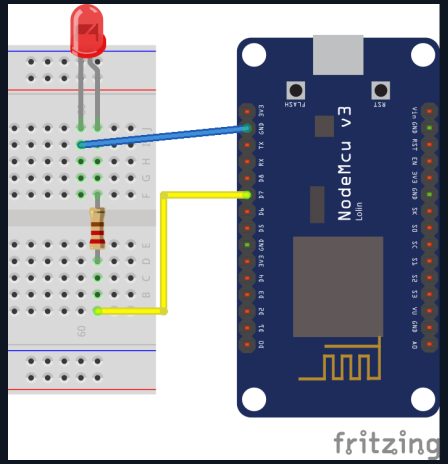




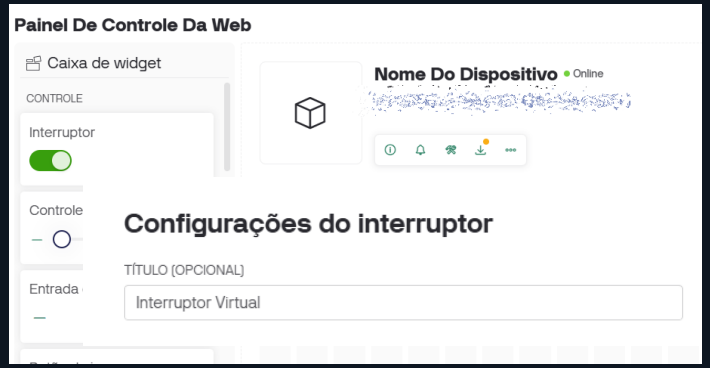


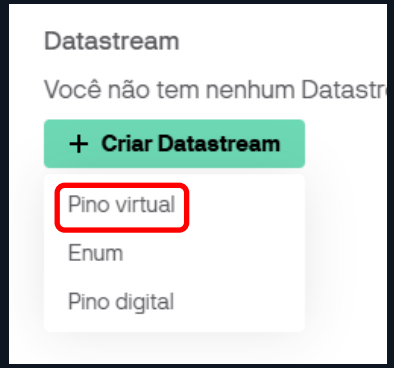


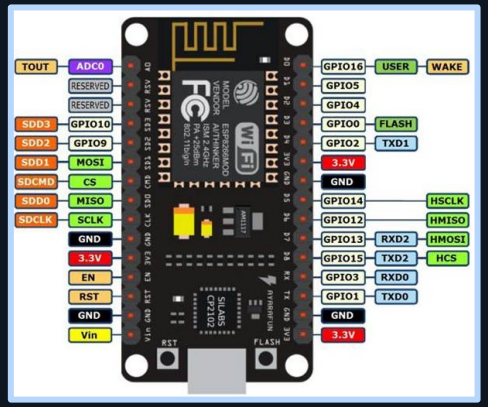


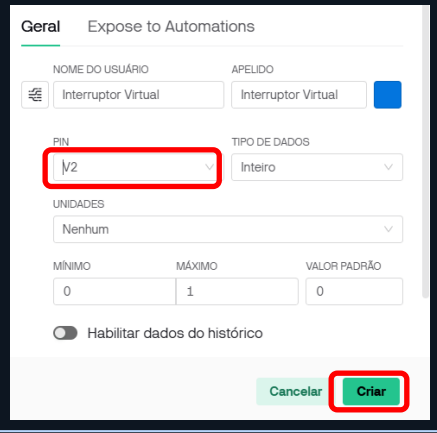


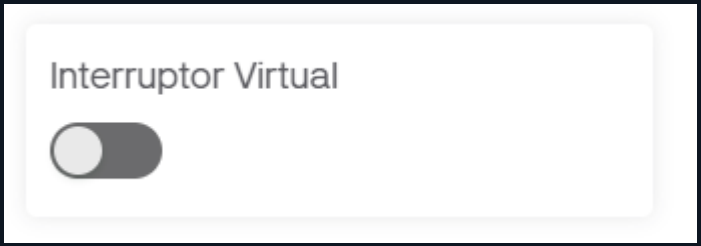
Blynk – Pino Virtual

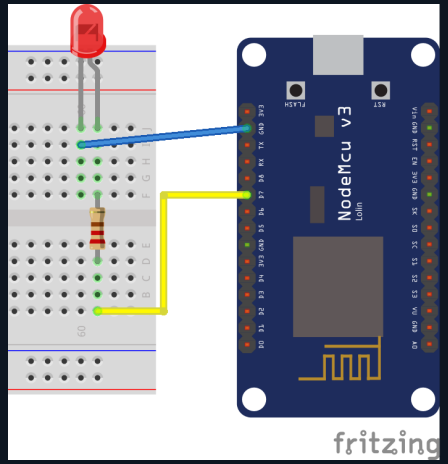




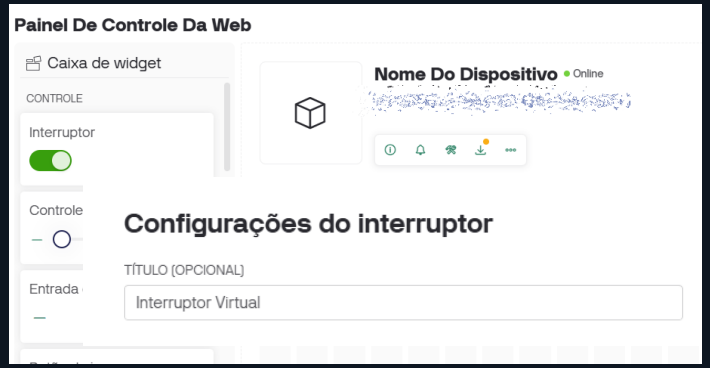


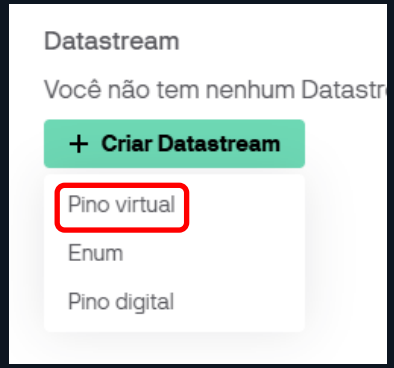


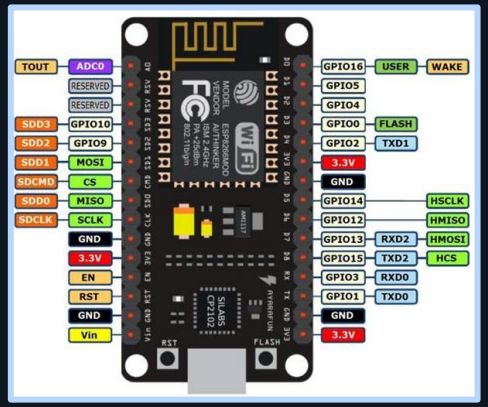


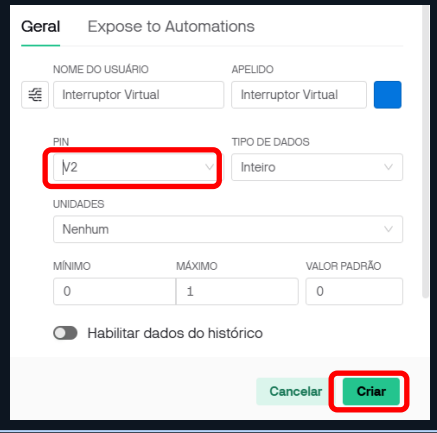


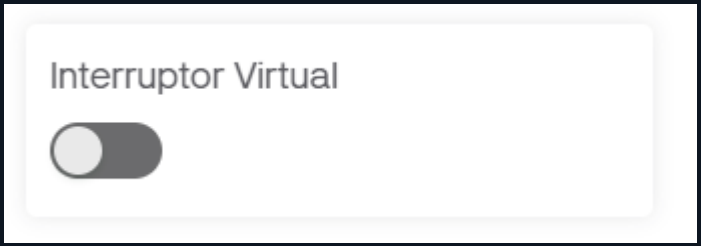
Blynk – Pino Virtual

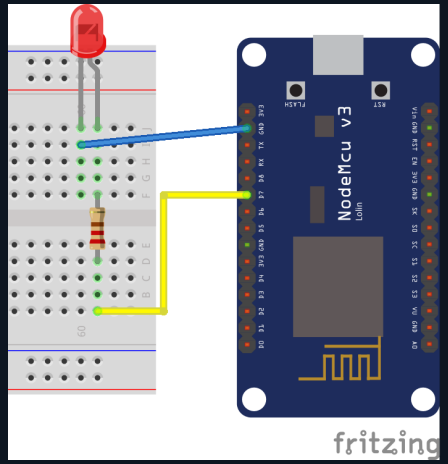






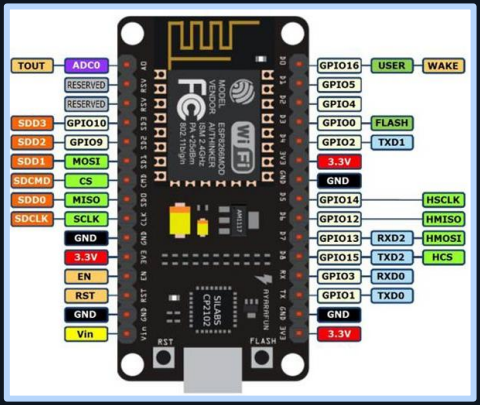


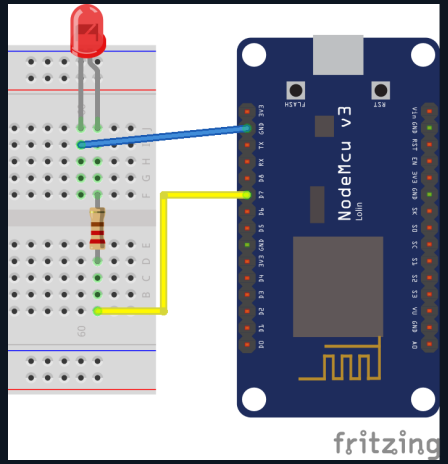




Blynk

```
#define BLYNK_PRINT Serial
#define BLYNK_TEMPLATE_ID “ID"
#define BLYNK_TEMPLATE_NAME “TEMP"
#define BLYNK_AUTH_TOKEN “TKN"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

int led=13;

char ssid[] = “REDE";
char pass[] = “SENHA";

void setup()
{
  Serial.begin(9600);
  pinMode(led, OUTPUT);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
}
```

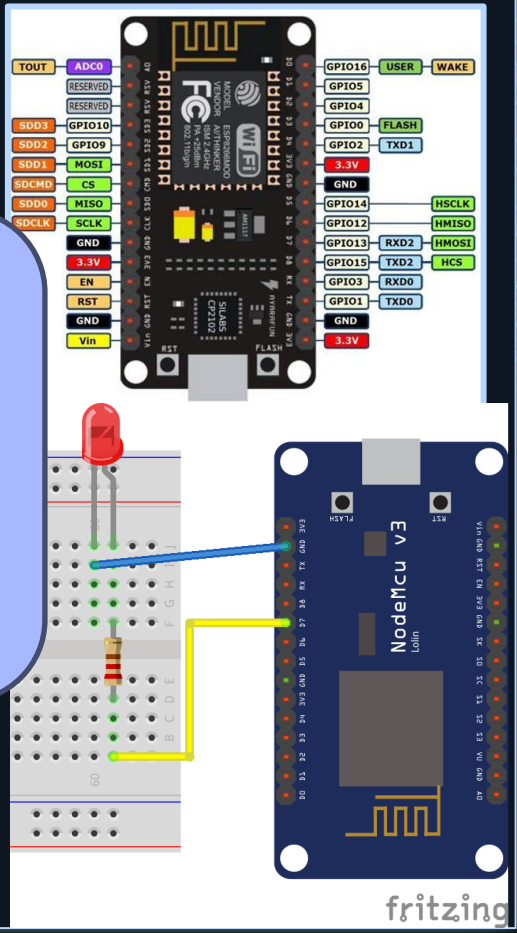




Blynk

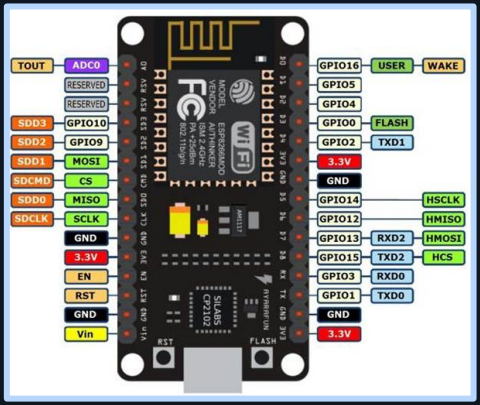
```
void loop()
{
  Blynk.run();
}

BLYNK_WRITE(V2) {
  int pinValue = param.asInt();
  digitalWrite(led, pinValue);
}
```

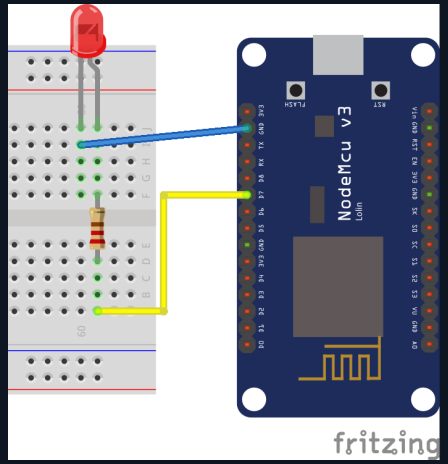
Para receber um valor do sistema virtual devemos criar uma Função.
Os parâmetros de conversão temos:
asInt() → inteiro
asFloat() → decimal
asDouble() → dupla precisão
asString() → cadeia de caractere
asChar() → ponteiro de caractere
asJson() → retorna um Json



Blynk

Método	Tipo de Retorno	Uso Comum
asInt()	Inteiro (int)	Controle de estados binários (1 ou 0).
asFloat()	Ponto flutuante (float)	Leituras de sensores analógicos.
asDouble()	Precisão dupla (double)	Cálculos precisos em aplicações avançadas.
asString()	String (String)	Texto para exibição ou mensagens.
asChar()	Ponteiro (char*)	Comunicação com APIs ou funções de C.
asJson()	JSON (String)	Enviar ou receber dados complexos.

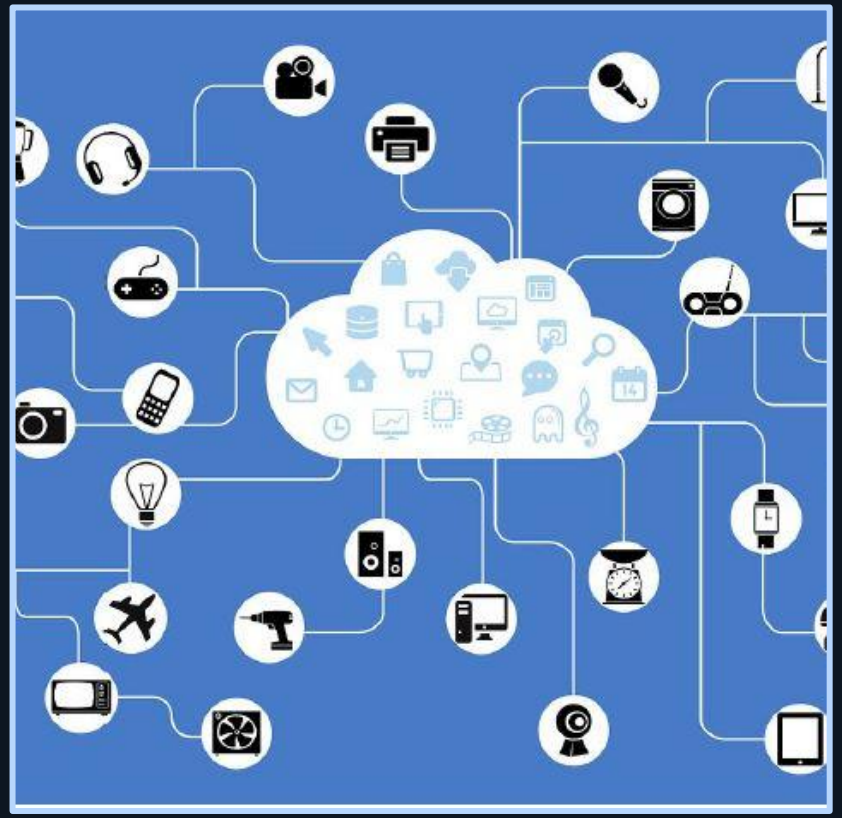




Blynk

Widgets Comuns Usando Pinos Virtuais

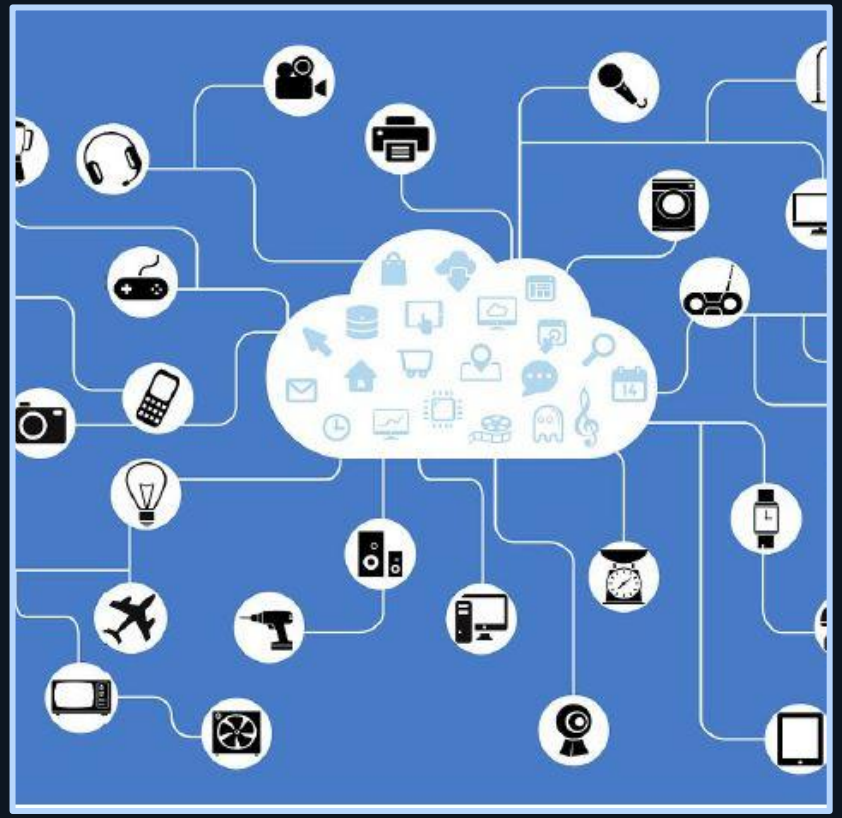
- **Botão:**
  - Liga/desliga dispositivos.
- **Slider:**
  - Controla valores analógicos.
- **Gráfico:**
  - Monitora dados em tempo real.
- **Gauge (Medidor):**
  - Exibe valores em forma de indicador.



Blynk

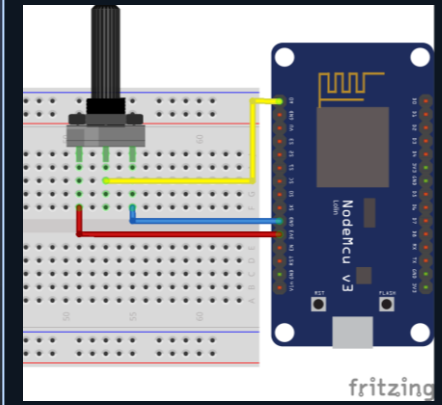
Boas Práticas com Pinos Virtuais

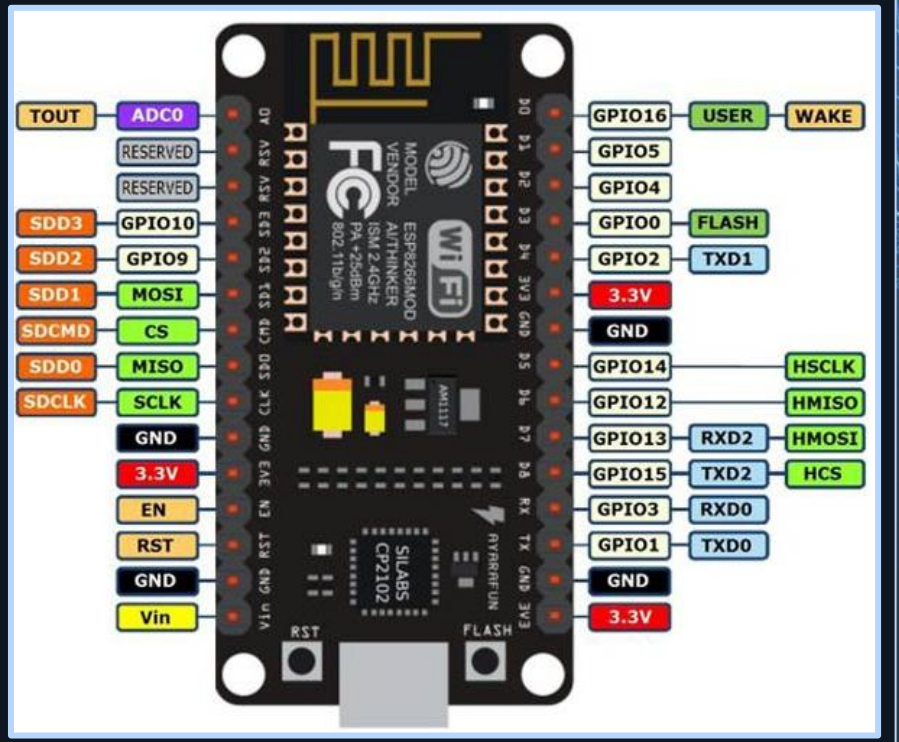
- Use nomes descritivos nos widgets para facilitar o entendimento (ex.: "Temperatura" para V2).
- Centralize a lógica no microcontrolador, mantendo o dashboard simples.
- Teste as interações entre pinos virtuais e físicos antes de implementar em projetos maiores.



Desafio 01

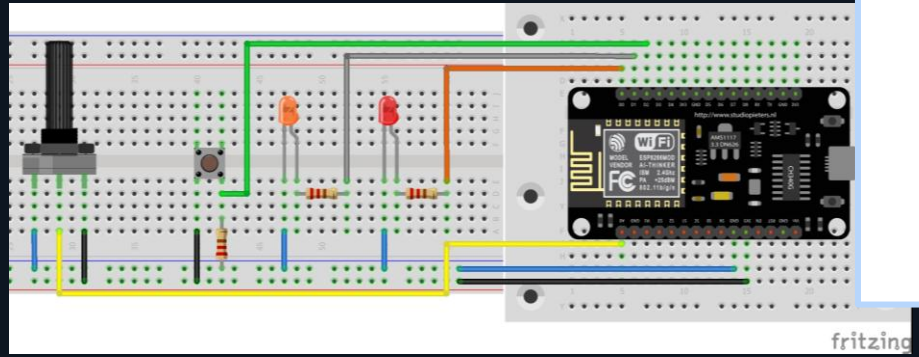
Vamos melhorar ainda mais:
Crie um programa que faça a leitura de 5 em 5 segundo e só mande a informação se o valor atual for maior (atual + 5) ou menor (atual - 5) que o valor anterior.

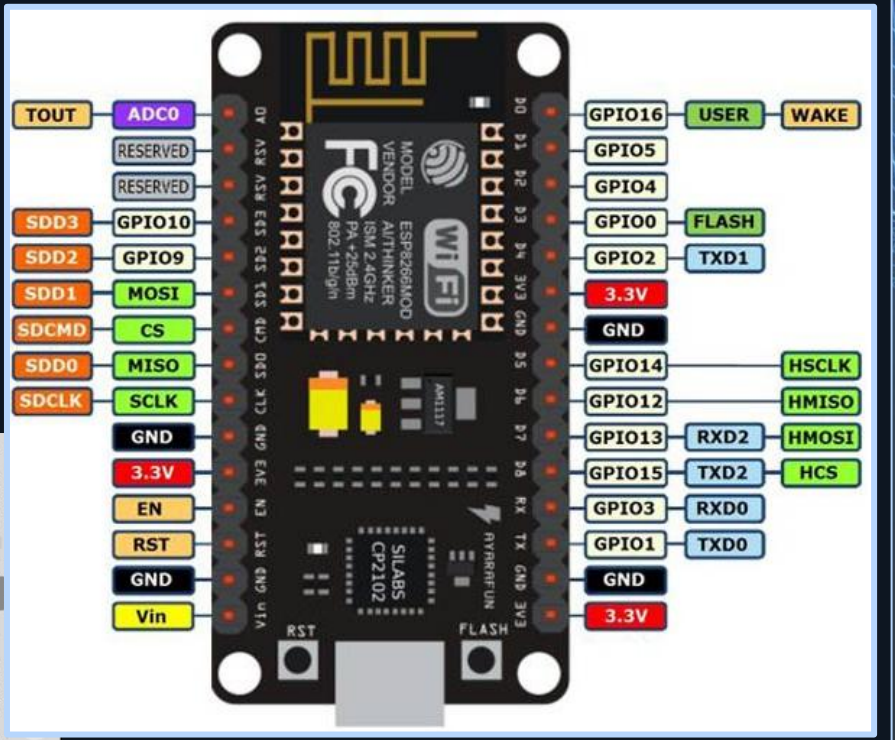




Desafio 02

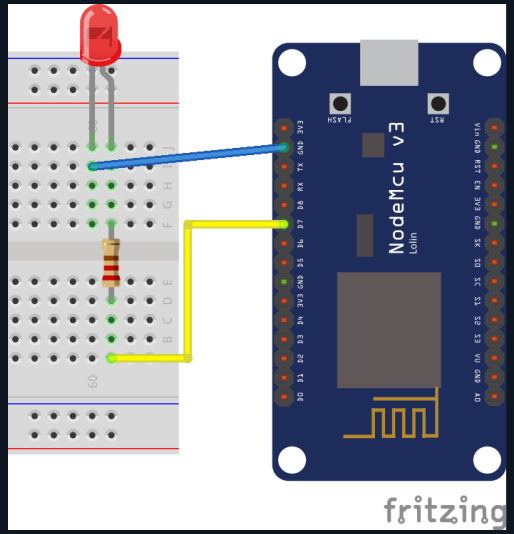
Vamos juntar tudo que aprendemos até aqui e incluir um botão com debounce de LIGA/DESLIGA um LED Virtual, e vamos colocar 2 LEDs um de cada cor.
Faça o código e o Dashboard desse código.

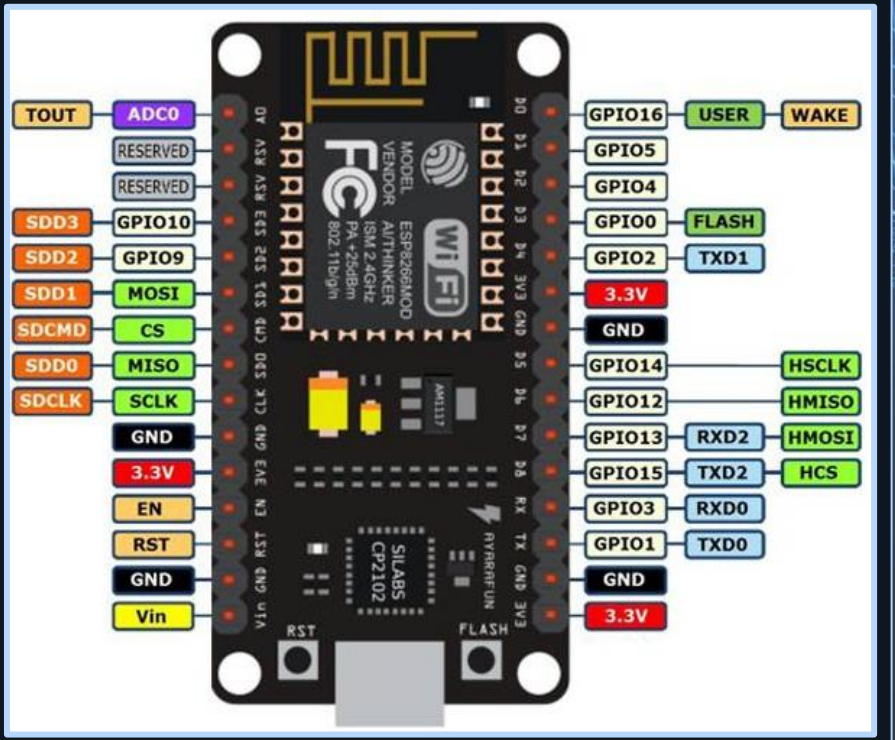




Desafio 03

Faça um código para controlar o LED por sistema PWM, coloque um slidebar.





Obrigado!